

TensorBinding.jl

Package Function Reference

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Overview

TensorBinding.jl is a Julia package for constructing and manipulating tight-binding Hamiltonians as Matrix Product Operators (MPOs) in the *quantics binary* (QTT) representation. A system of $N = 2^L$ sites is encoded in L qubit sites: site index $n \in \{0, \dots, N-1\}$ is identified with the binary string $b_1 b_2 \dots b_L$. This representation makes the bond dimension of many physically relevant Hamiltonians small (typically ≤ 10), enabling compression via Quantics Tensor Cross Interpolation (QTCI).

Key capabilities:

- MPO construction for 1D and 2D lattices (square, triangular, triangular Bravais, honeycomb, kagomé, Lieb, dice) via shift operators and QTCI. Sublattice-aware models carry an explicit sublattice index and exact real-space positions.
- Generic n th-nearest-neighbor hopping via `add_hopping_2D!`: uniform, direction-dependent, site-dependent, or fully position+direction-dependent amplitude functions on any 2D geometry.
- Bilayer, multilayer, twisted, and Bernal-stacked Hamiltonians.
- Spin- $\frac{1}{2}$ and Nambu (BdG) extensions via a prepend/postpend-core pattern, with auxiliary DOF placeable at front (`:pre`) or back (`:post`).
- Kernel Polynomial Method (KPM) for density of states, density matrices, spectral functions, and Green's functions.
- Density-matrix purification (McWeeny, SP2) with `TBHamiltonian`-aware API.
- Quantum Fourier Transform conjugation for band structure $A(k, \omega)$, supporting spinful, BdG, and layered Hamiltonians via `aux_proj`.
- Topological invariants: real-space Chern marker and winding-number density via quenched or flat position operators, compatible with any lattice geometry.
- DMRG ground state and spectral DMRG.
- Real-time evolution: TDVP (fixed and time-dependent H), propagator MPO.
- Random Phase Approximation (RPA) for susceptibility.
- Self-consistent mean-field (Hubbard model).

Site encoding conventions:

- *Position qubits*: L sites, site i (0-indexed) $\leftrightarrow$ binary string $b_1 \dots b_L$, big-endian.
- *Extended systems*: auxiliary indices can be *prepended* (`:pre`, default) or *postpended* (`:post`). For a spinful bilayer BdG system with `:pre` the site chain is `[nambu_s, spin_s, layer_s, pos_qubits...]`.
- *2D lattices*: row-major flattening, linear index $i = i_x + i_y \cdot 2^{L_x}$, $i_x \in \{0, \dots, 2^{L_x} - 1\}$, $i_y \in \{0, \dots, 2^{L_y} - 1\}$.
- *Honeycomb (legacy "hex_2d")*: intra-row bonds pair even and odd i_x sites; two inter-row bond types controlled by sublattice masks. Requires $L = L_x + L_y$ total qubits.
- *Sublattice-explicit models* ("`honeycomb`", "`honeycomb_nnn`", "`kagome`", "`lieb`", "`dice`"): use a triangular Bravais basis of $2^{L_x} \times 2^{L_y}$ unit cells ($L = L_x + L_y$ qubits) plus one postpended sublattice index of dimension $n_{\text{sub}} \in \{2, 3\}$. Total physical sites = $n_{\text{sub}} \cdot 2^L$; H.N counts unit cells.

Central Data Structure: TBHamiltonian

Defined in `src/core/TBSystem.jl`.

Field	Type	Description
<code>L</code>	<code>Int</code>	Number of position qubits; $N = 2^L$ sites.
<code>N</code>	<code>Int</code>	Number of physical sites per spin/Nambu/layer sector.
<code>sites</code>	<code>Vector{<:Index}</code>	All ITensor site indices (includes auxiliary indices if present).
<code>mpo</code>	<code>MPO</code>	The accumulated Hamiltonian MPO.
<code>geometry</code>	<code>Function</code> or <code>Nothing</code>	Callable <code>i -> pos_vec</code> (1-indexed); returns real-space position of physical atom <code>i</code> .
<code>geometry_uc</code>	<code>Function</code> or <code>Nothing</code>	Callable <code>i -> pos_vec</code> returning the Bravais unit-cell position, <i>identical</i> for all sublattice atoms in the same cell. Set for sublattice-aware models (<code>"honeycomb"</code> , <code>"honeycomb_nnn"</code> , <code>"kagome"</code> , <code>"lieb"</code> , <code>"dice"</code>); <code>nothing</code> for single-atom geometries.
<code>scale</code>	<code>Float64</code>	Energy half-bandwidth; <code>H/scale</code> has spectrum $\subseteq [-1, 1]$. Set to 0 for lazy DMRG estimation.
<code>center</code>	<code>Float64</code>	Spectral center $(E_{\max} + E_{\min})/2$. 0 for particle-hole symmetric spectra.
<code>spin_s</code>	<code>Index</code> or <code>Nothing</code>	Spin- $\frac{1}{2}$ index (set by <code>add_spin!</code>).
<code>nambu_s</code>	<code>Index</code> or <code>Nothing</code>	Nambu index (set by <code>add_superconductivity!</code>).
<code>layer_s</code>	<code>Index</code> or <code>Nothing</code>	Layer index (set by bilayer/multilayer constructors).
<code>sublattice_s</code>	<code>Index</code> or <code>Nothing</code>	Sublattice index postpended by sublattice-aware constructors (<code>honeycomb</code> , <code>kagomé</code> , <code>Lieb</code> , <code>dice</code>).
<code>aux_side</code>	<code>Symbol</code>	<code>:pre</code> (aux at front, default) or <code>:post</code> (aux at back).
<code>_tn_cache</code>	<code>Vector{MPO}</code> or <code>Nothing</code>	Cached MPO Chebyshev list $T_0 \dots T_N$ (mode <code>:mpo</code>).
<code>_tn_mps_cache</code>	<code>Vector{MPS}</code> or <code>Nothing</code>	Cached MPS Chebyshev list (mode <code>:mps</code>); set by <code>KPM_Tn</code> with <code>mode=:mps</code> .
<code>_tn_Ncheb</code>	<code>Int</code>	Order N of cached Chebyshev list (shared by both caches).
<code>_density_cache</code>	<code>MPO</code> or <code>Nothing</code>	Cached density matrix from purification.
<code>interaction_mpo</code>	<code>MPO</code> or <code>Nothing</code>	Hartree/CDW interaction kernel set by <code>add_interaction!</code> ; used by <code>get_scf(H, channel)</code> .
<code>fock_mpo</code>	<code>MPO</code> or <code>Nothing</code>	Fock/exchange interaction kernel set by <code>add_interaction!(H, V; channel=:fock)</code> .
<code>Lx</code>	<code>Int</code> or <code>Nothing</code>	Number of x -direction qubits for 2D geometries ($L_y = L - L_x$); <code>nothing</code> for 1D/non-rectangular geometries.

Do not construct directly; use `get_Hamiltonian` or a layer/multilayer constructor. Caches are invalidated automatically by all `add_*` functions and `truncate!`. The `center` field is set together with `scale` during spectral-bounds estimation and is needed to correctly rescale energies: $\tilde{\omega} = (\omega - \text{center})/\text{scale}$.

Hamiltonian Construction (TBSystem.jl)

Constructor

Signature	Description
<code>get_Hamiltonian(geometry, params; L, scale=nothing, tol=1e-8, maxdim=15, ref_sites=nothing, kwargs...)</code>	Build a <code>TBHamiltonian</code> from a named model. <code>geometry</code> is a string key; <code>params</code> holds model parameters (see below). <code>ref_sites</code> (optional): if supplied, the constructed MPO's site indices are replaced in-place with <code>ref_sites</code> via <code>fix_sites</code> , so that multiple Hamiltonians built independently share identical <code>Index</code> objects. Returns a <code>TBHamiltonian</code> .

Supported geometry strings:

geometry	params	Notes
"chain_1d"	hopping t	1D NN chain, direct MPO (no QTCI).
"square_2d"	hopping t	2D square, $2^{L_x} \times 2^{L_y}$ sites; Lx/Ly kwargs.
"hex_2d"	hopping t	2D honeycomb.
"triangular_2d"	hopping t	2D triangular.
"ssh"	(t, d)	SSH chain: hopping $t \pm d$ on alternating bonds.
"ssh_sublattice"	(t, d)	SSH chain with an explicit 2-dim SSH sublattice index (H.sublattice_s, aux_side=:post) instead of alternating bond amplitudes.
"aah"	(V, ϕ, t)	Aubry–André–Harper quasicrystal.
"uniform"	t	Uniform 1D with optional on-site v .
"haldane"	(t_2, ϕ, M) NamedTuple	Haldane model; requires rs ($N \times 2$ position matrix).
"chernhex"	(t, t_2, ms)	Haldane-type Chern insulator on honeycomb.
"chern8"	(V, t)	8-fold Chern mosaic on square lattice.
"qc2dsquare"	t	Quasicrystal-modulated square lattice.
"honeycomb"	$(t,)$ NamedTuple	Honeycomb with explicit 2-atom sublattice index (postpended); requires Lx/Ly kwargs.
"honeycomb_nnn"	(t, t_2) NamedTuple	Honeycomb with NN hopping t and NNN same-sublattice hopping t_2 .
"kagome"	$(t,)$ NamedTuple	Kagomé lattice, 3-atom unit cell; Lx/Ly required.
"lieb"	$(t,)$ NamedTuple	Lieb lattice, 3-atom unit cell; flat band at $E = 0$.
"dice"	$(t,)$ NamedTuple	Dice (T3) lattice, 3-atom unit cell (hub A, rim B/C).
"triangular_bravais"	hopping t	Bravais triangular lattice: $\mathbf{a}_1 = (1, 0)$, $\mathbf{a}_2 = (\frac{1}{2}, \frac{\sqrt{3}}{2})$, three clean bond types; Lx/Ly required.
"custom"	$f(i, j)$	Arbitrary hopping matrix; scale required.

Sublattice-aware geometries ("honeycomb", "honeycomb_nnn", "kagome", "lieb", "dice") postpend a sublattice_s index of dimension n_{sub} (2 or 3) to the site chain. The physical site count becomes $n_{\text{sub}} \cdot 2^{L_x+L_y}$; H.geometry maps the enlarged 1-indexed site number to a real-space 2D position, while H.geometry_uc maps it to the shared Bravais unit-cell position (same value for all n_{sub} atoms in a cell). H.Lx is set to L_x for these geometries and for all 2D presets ("square_2d", "hex_2d", "triangular_2d", "triangular_bravais", "chern8", "chernhex", "qc2dsquare"), and is nothing for 1D/custom geometries; add_hopping! dispatches on H.Lx to redirect to add_hopping_2D!.

Cache and Truncation Helpers

Signature	Description
<code>_invalidate_cache!(H) → H</code>	Clear all cached results (<code>_tn_cache</code> , <code>_tn_mps_cache</code> , <code>_density_cache</code> , <code>scale/center</code>). Called automatically by all <code>add_*</code> functions.
<code>truncate!(H; cutoff=1e-10, maxdim=nothing) → H</code>	Truncate <code>H.mpo</code> in-place; invalidates all caches.

Incremental Hamiltonian Modification

All functions below mutate `H` in place and invalidate the Chebyshev cache. `add_hopping!` and `add_onsite!` must be called before `add_spin!` or `add_superconductivity!`.

The optional `position` keyword (default: inherits `H.aux_side`) controls whether auxiliary indices are prepended (`:pre`) or postpended (`:post`).

Signature	Description
<code>add_hopping!(H, f; nn=1, boundary=:open, bc=nothing, maxdim=15, tol=1e-8, type=ComplexF64, apply_kwargs=NamedTuple(), sublat=nothing, sublat_from=nothing, sublat_to=nothing)</code>	Add a hopping term to <code>H.mpo</code> . The mode of <code>f</code> is selected by type: <code>f::Number</code> gives a uniform <code>nn</code> -th-neighbour hopping (shell <code>nn</code>); <code>f(i)</code> (1-arg Function) gives a spatially varying amplitude per site, QTCI-compressed; <code>f(i,j)</code> (2-arg Function) gives a full $N \times N$ matrix via QTCI (<code>nn</code> ignored). <code>boundary</code> (or its alias <code>bc</code>) is <code>:open</code> or <code>:periodic</code> , controlling the shift operators used for <code>nn</code> -th-neighbour terms. <code>apply_kwargs</code> is forwarded to the underlying <code>apply</code> call (e.g. <code>(maxdim=50, cutoff=1e-10,)</code>). If <code>H.Lx</code> is set (2D Hamiltonian), dispatches to <code>add_hopping_2D!</code> instead (sublattice keywords are not supported in that case). Without sublattice keywords, errors if any auxiliary DOF index is already present. Invalidates all caches.
<code>add_hopping!(H, f; sublat=k, ...)</code>	<i>Intra-sublattice</i> : requires <code>H.sublattice_s</code> . Restricts the full kinetic term to sublattice k by tensoring with the projector $ k\rangle\langle k $. Works with all three <code>f</code> modes; must be called before <code>add_spin!</code> / <code>add_superconductivity!</code> . <i>e.g.</i> <code>add_hopping!(H, -0.1; sublat=1, nn=2)</code> (NNN within sublattice A).
<code>add_hopping!(H, f; sublat_from=k, sublat_to=l, nn=1, ...)</code>	<i>Inter-sublattice</i> : requires <code>H.sublattice_s</code> and a scalar <code>f</code> . Adds $f \cdot K_u^{nn} \otimes l\rangle\langle k + \bar{f} \cdot K_d^{nn} \otimes k\rangle\langle l $ (Hermitian). <code>nn=0</code> gives an intra-cell bond (no position shift) — useful for correcting individual bond amplitudes in an existing sublattice Hamiltonian. <i>e.g.</i> <code>add_hopping!(H, -0.3; sublat_from=1, sublat_to=2, nn=1)</code> ($A \leftrightarrow B$ NN inter-cell).

<code>add_onsite!(H, f; layer=nothing, sublat=nothing, Lx=nothing, tol=1e-8, maxdim=nothing)</code>	Add a diagonal on-site potential to H. $f::\text{Number}$ gives a uniform constant ($f \cdot \text{Id}$, no QTCI); $f(\mathbf{n})$ (0-indexed unit-cell index $n \in \{0, \dots, N-1\}$) gives a spatially varying 1D potential; $f(\mathbf{i}_x, \mathbf{i}_y)$ (0-indexed 2D coordinates, $i_x = n \bmod N_x$, $i_y = n \div N_x$) requires the <code>Lx</code> keyword so $N_x = 2^{L_x}$. <code>sublat=k</code> restricts the potential to sublattice k via $k\rangle\langle k$ (requires <code>H.sublattice_s</code>); e.g. a Semenoff mass on honeycomb: <code>add_onsite!(H, +M; sublat=1),</code> <code>add_onsite!(H, -M; sublat=2).</code> <code>layer=k</code> (or a list of layer indices) restricts to the selected layer(s) (requires <code>H.layer_s</code>); <code>layer=nothing</code> (default) applies to all layers.
<code>add_spin!(H; cutoff=1e-8, maxdim=200, position=:pre)</code> <code>add_zeeman!(H, h; direction=:z, tol=1e-8, maxdim=200, position)</code>	Prepend (or postpend) a spin-$\frac{1}{2}$ index: $\mathbf{I}_{\text{spin}} \otimes H$. No-op if already spinful. Add Zeeman coupling $h \cdot S_\alpha$. h is a Number or function $h(i)$. <code>direction</code> $\in \{x, y, z\}$. Calls <code>add_spin!</code> automatically. If BdG is already applied, wraps in τ_z.
<code>add_superconductivity!(H, Δ; type=:swave, tol=1e-8, maxdim=200, position)</code>	BdG extension: prepend (or postpend) a Nambu index, with $H_{\text{BdG}} = \tau_z \otimes H_{\text{kin}} + \tau_+ \otimes H_{\text{pair}} + \tau_- \otimes H_{\text{pair}}^\dagger$. <code>type=:pwave</code> (spinless only, Δ a Number): antisymmetric NN pairing $\Delta(K_u - K_d)$ (Kitaev chain). <code>type=:swave</code> (default): for spinful H, singlet pairing $(i\sigma_y)_{\text{spin}} \otimes \Delta(r)$ with Δ a Number or 1-arg function $\Delta(i)$; for spinless H, on-site pairing is forbidden by Fermi antisymmetry, so a Number Δ auto-redirects to <code>type=:pwave</code> (printing an informational message) and a Function Δ errors (use <code>type=:custom</code> instead). <code>type=:custom</code>: arbitrary pairing $\Delta(i, j)$ via TCI (<code>pairing2MPO</code>). Errors if BdG has already been applied.
<code>add_soc!(H, λ; type=:rashba, direction=:z, tol=1e-8, maxdim=200, position)</code>	Spin-orbit coupling. Calls <code>add_spin!</code> automatically if needed. <code>type=:rashba</code>: NN Rashba $\lambda(i\sigma_y \otimes K_u - i\sigma_y \otimes K_d)$ (scalar λ only); breaks SU(2) spin symmetry while preserving time-reversal. <code>type=:ising</code>: diagonal $\lambda(i) \cdot S_z$ (Kane–Mele-type), λ a Number or 1-arg function. <code>type=:custom</code>: position-space MPO built from λ (Number, 1-arg function $\lambda(i)$, or 2-arg function $\lambda(i, j)$ via TCI), tensored with the spin operator given by <code>direction</code> ($\in \{x, y, z\}$). For Hermiticity, $\lambda(i, j) = \lambda(j, i)$ (automatic for diagonal/real-symmetric λ).

Typical workflow:

```
H = get_Hamiltonian("square_2d", 1.0; L=8)
```

```

add_onsite!(H, _ -> -0.5)           # chemical potential
add_zeeman!(H, 0.1)                 # Zeeman splits spin sectors
add_soc!(H, 0.05)                   # Rashba SOC
add_superconductivity!(H, 0.2)      # s-wave BdG
# H.sites = [nambu_s, spin_s, pos_qubits...] (default :pre stacking)

```

Interaction Storage

Signature	Description
<code>add_interaction!(H, V; channel=:hartree, type=Float64, tol=1e-8, kwargs...) → H</code>	Store a pre-built interaction MPO in H for later use with <code>get_scf(H, channel)</code>. V can be an MPO (stored directly), a <code>Number</code> (scalar coupling $V \cdot \text{Id}$), a 1-arg <code>Function</code> $V(i)$ (0-indexed on-site potential, stored as a diagonal MPO), or a 2-arg <code>Function</code> $V(i, j)$ (0-indexed interaction kernel, stored as a full 2D MPO via <code>get_mpo</code>). <code>channel=:hartree</code> (or <code>:default</code>) stores into <code>H.interaction_mpo</code> (used by the <code>:cdw</code> and <code>:magnetic</code> SCF channels); <code>channel=:fock</code> (or <code>:exchange</code>) stores into <code>H.fock_mpo</code>. Does not invalidate the Chebyshev/density caches.

Geometry Helpers

Signature	Description
<code>honeycomb_positions(L; Lx=L div 2) → Matrix{Float64}</code>	Return $N \times 2$ real-space positions for a honeycomb patch ($N = 2^L$).
<code>square_positions(L; Lx=L div 2) → Matrix{Float64}</code>	Return $N \times 2$ real-space positions for a square lattice patch.
<code>triangular_positions(L; Lx=L div 2) → Matrix{Float64}</code>	Return $N \times 2$ real-space positions for a triangular lattice patch. Odd rows offset by 0.5 in x ; $y = i_y \cdot \sqrt{3}/2$.
<code>triangular_bravais_positions(L; Lx=L div 2) → Matrix{Float64}</code>	Positions for Bravais triangular lattice: $\mathbf{r}(i_x, i_y) = i_x \mathbf{a}_1 + i_y \mathbf{a}_2$ with $\mathbf{a}_1 = (1, 0)$, $\mathbf{a}_2 = (\frac{1}{2}, \frac{\sqrt{3}}{2})$.
<code>honeycomb_sublattice_positions(Lx, Ly) → Matrix{Float64}</code>	$2N_{\text{uc}} \times 2$ positions for honeycomb with explicit sublattice: A at $(i_x + 0.5 i_y, i_y \sqrt{3}/2)$, B at $(\dots + 0.5, \dots + \sqrt{3}/6)$.
<code>kagome_positions(Lx, Ly) → Matrix{Float64}</code>	$3N_{\text{uc}} \times 2$ positions for the kagomé lattice (A, B, C sublattices).
<code>lieb_positions(Lx, Ly) → Matrix{Float64}</code>	$3N_{\text{uc}} \times 2$ positions for the Lieb lattice (A, B, C sublattices).
<code>dice_positions(Lx, Ly) → Matrix{Float64}</code>	$3N_{\text{uc}} \times 2$ positions for the dice (T3) lattice (hub A, rim B/C).
<code>lattice_positions(lattice, Lx, Ly; angle_deg=0.0) → Matrix{Float64}</code>	Positions for <code>:square</code> , <code>:triangular</code> , or <code>:honeycomb</code> , optionally rotated by <code>angle_deg</code> degrees about the centroid. (<code>twisted.tk.jl</code>)

<code>central_index(geom, N) → Int</code>	1-indexed site whose position is closest to the geometric centroid. <code>geom</code> is a callable <code>i → pos_vec</code> . Errors if <code>H.geometry</code> is <code>nothing</code> .
<code>central_index(H) → Int</code>	
<code>_pos_sites(H) → Vector{<:Index}</code>	The L position-qubit indices, obtained by filtering <code>H.sites</code> to remove <code>H.spin_s</code> , <code>H.nambu_s</code> , <code>H.layer_s</code> , and <code>H.sublattice_s</code> (whichever are set), regardless of whether they sit at the front or back of <code>H.sites</code> .

Low-Level MPO Builders (Hamiltonian.jl, 2D_lattice.jl)

These functions build MPOs directly on a provided `sites` vector.

1D Kinetic Operators

Signature	Description
<code>kinetic_1d_nn(L, sites; boundary=:open, bc=nothing) → MPO</code>	Uniform NN hopping $\sum_i (i+1\rangle\langle i + \text{h.c.})$ via the shared <code>shift_pair_mpos</code> shift/adjoint pair. <code>boundary=:open</code> (default) is the open-chain convention; <code>boundary=:periodic</code> (or <code>bc=:pbc</code>) adds the wrap-around bond (<code>cyclic=true</code>).
<code>kinetic_1d_nn_custom(L, sites, hopping; boundary=:open, bc=nothing) → MPO</code>	Spatially varying NN hopping; <code>hopping</code> is a diagonal MPO of amplitudes, combined via <code>shift_hopping_mpo</code> . Same boundary/bc convention as <code>kinetic_1d_nn</code> .
<code>kineticNNN(L, sites, hopping, nn; apply_kwargs=NamedTuple(), boundary=:open, bc=nothing) → MPO</code>	1D chain with n -th-nearest-neighbour hopping and a spatially varying amplitude encoded as the diagonal MPO <code>hopping</code> , via the shared <code>shift_hopping_mpo</code> helper $(\text{hopping} \cdot \text{shift}(nn) + \text{shift}(nn)^\dagger \cdot \overline{\text{hopping}})$. <code>nn</code> ≥ 1 ; <code>boundary=:open</code> (default) or <code>:periodic/bc=:pbc</code> for a cyclic shift.
<code>generate_kin_u(sites, N) → MPO</code>	Shift MPO $ n+1\rangle\langle n $ (binary increment / up-shift).
<code>generate_kin_d(sites, N) → MPO</code>	Shift MPO $ n-1\rangle\langle n $ (binary decrement / down-shift).

2D Kinetic Operators

Signature	Description
<code>intrachain_hopping(L_chain, N, sites; hopping=Id, t=1) → MPO</code>	NN x -direction hopping on a 2D lattice with row-break mask applied.
<code>interchain_hopping_square(L_chain, N, sites; hopping=Id, t=1) → MPO</code>	NN y -direction hopping on a square lattice.
<code>interchain_hopping_square_2nd_plus(L_chain, N, sites; hopping=Id, t2=1) → MPO</code>	NN hopping along $(+x, +y)$ diagonal (shift $+L_{\text{chain}} + 1$); row-break mask prevents wrap-around.
<code>interchain_hopping_square_2nd_minus(L_chain, N, sites; hopping=Id, t2=1) → MPO</code>	NN hopping along $(+x, -y)$ diagonal (shift $-(L_{\text{chain}} - 1)$).

<code>interchain_hopping_triangle(L_chain, N, sites) → MPO</code>	Both diagonal inter-row bonds on a triangular lattice (SW↗NE and SE↖NW).
<code>interchain_hopping_honeycomb(L_chain, N, sites) → MPO</code>	Inter-row hopping for a honeycomb lattice. Two inequivalent bonds, each with forward and backward terms: upper bond (shift $+L_{\text{chain}} + 1$) masked by <code>odd_template × odd_skeleton</code> (odd- i_x columns, even rows); lower bond (shift $+L_{\text{chain}} - 1$) masked by <code>even_template × even_skeleton</code> (odd- i_x columns excluding $i_x=1$, odd rows).
<code>kineticintra2DNNN(Lx, Ly, sites, hopping, nn; apply_kwargs) → MPO</code>	Long-range intra-row hopping on $2^{L_x} \times 2^{L_y}$ square lattice. Row-boundary wrap suppressed by <code>_row_break_mpo</code> .
<code>kineticintra2DNNhex(Lx, Ly, sites, hopping, nn; apply_kwargs) → MPO</code>	Intra-row hopping for honeycomb; applies checkerboard A/B mask.
<code>kineticinterNNNSWNE(Lx, Ly, sites, hopping, nn; apply_kwargs) → MPO</code>	SW↗NE diagonal inter-row hopping (square lattice, <code>:xplus</code> mask).
<code>kineticinterNNNSENW(Lx, Ly, sites, hopping, nn; apply_kwargs) → MPO</code>	SE↖NW diagonal inter-row hopping (<code>:xplain</code> mask).
<code>kineticinterNNNtriSWNE(Lx, Ly, sites, hopping, nn; apply_kwargs) → MPO</code>	SW↗NE diagonal for triangular lattice; adds even-row selection mask.
<code>kineticinterNNNtriSEnw(Lx, Ly, sites, hopping, nn; apply_kwargs) → MPO</code>	SE↖NW diagonal for triangular lattice; adds odd-row selection mask.
<code>kineticinterNNNtri.bravais_diag(Lx, Ly, sites, hopping; apply_kwargs) → MPO</code>	Third-bond hopping for Bravais triangular lattice ($\Delta i_x=+1, \Delta i_y=-1$), linear shift $-(N_x - 1)$; row x -wrap at $i_x=N_x - 1$ suppressed by <code>_row_break_mpo(:xplus)</code> .

Diagonal Mask MPOs

Used internally as connectivity filters; all return diagonal 0/1 MPOs.

Signature	Description
<code>_row_break_mpo(Lx, Ly, sites; which::Symbol) → MPO</code>	Mask 0 at row boundaries. <code>which=:xplus</code> : last column; <code>:xplain</code> : first column.
<code>_row_select_mpo(_, Ly, sites; keep=:even) → MPO</code>	Retain only even or odd rows (<code>keep=:even</code> or <code>:odd</code> , 0-based).
<code>_col_select_mpo(Lx, Ly, sites; keep=:even) → MPO</code>	Retain only even or odd columns.
<code>_row_checker_mpo(Lx, Ly, sites) → MPO</code>	Checkerboard: 1 where $(i_x + i_y)$ is even, 0 otherwise.
<code>skeleton(L_chain, N, sites) → MPO</code>	0 where $i_x = 1$ (second column, 0-indexed), 1 elsewhere. Used internally for triangular wrap-around suppression.
<code>odd_template(_, N, sites) → MPO</code>	1 where i_x is odd (LSB of $i_x = 1$). Selects the A-sublattice columns for the honeycomb inter-row bonds.

<code>even_template(L_chain, N, sites) → MPO</code>	1 where i_x is odd and $i_x \neq 1$, i.e. $\text{proj}_{\text{odd } i_x} - \text{proj}_{i_x=1}$. Selects the B-sublattice columns, excluding the boundary column that wraps into the next row.
<code>odd_skeleton(L_chain, N, sites) → MPO</code>	1 where i_y is even (0-based, LSB of $i_y = 0$). Selects even rows for the upper honeycomb inter-row bond.
<code>even_skeleton(L_chain, N, sites) → MPO</code>	1 where i_y is odd (0-based, LSB of $i_y = 1$). Selects odd rows for the lower honeycomb inter-row bond.

QTCI-Based Builders

Signature	Description
<code>hopping2MPO(f, N, sites; tol=1e-8, initial_positions=[], type=Float64, unfoldingscheme=:interleaved) → MPO</code>	Compress $N \times N$ matrix $H[i, j] = f(i, j)$ into an MPO via QTCI. <code>unfoldingscheme=:interleaved</code> (default) or <code>:fused</code> .
<code>qtc_matrix_to_MPO(A_fun, L, sites; tol=1e-8, type=Float64) → MPO</code>	Like <code>hopping2MPO</code> for $2^L \times 2^L$ matrix functions; applies extra truncation at <code>maxdim=20</code> .
<code>get_diagonal_mpo(L, sites, f; type=Float64) → MPO</code>	Diagonal MPO with entry $f(x)$ at site $x \in \{1, \dots, 2^L\}$ (1-indexed). Compressed via 1D QTCI.
<code>qtt_mpo(L, xvals, sites, func; tol_quantics=1e-8, maxbonddim_quantics=50) → MPO</code>	Compress scalar function <code>func(x)</code> on grid <code>xvals</code> (e.g. <code>0:2^L-1</code>) into a diagonal L -site MPO via QTCI. Used throughout the preset Hamiltonians below to encode on-site potentials and hopping amplitudes; see also <code>get_diagonal_mpo</code> .
<code>compose_power(base, nn; side=:right, apply_kwargs) → MPO</code>	base^{nn} by exponentiation-by-squaring in $O(\log n)$ applies. Returns identity if <code>nn=0</code> .

Preset Model Hamiltonians (direct MPO)

Function	Description
<code>HUniform(L, t; v=1e-6, tol_quantics=1e-8, maxbonddim_quantics=10, nn=1)</code>	Uniform 1D chain with optional on-site potential <code>v</code> . <code>v=0</code> triggers a warning (constant functions can fail QTCI).
<code>HSSH(L, t, d; tol_quantics=1e-8, maxbonddim_quantics=10, nn=1)</code>	SSH dimerized chain: hopping $t \pm d$ on alternating bonds.
<code>HAAH(L, V, phi, t; b=(1+√5)/2, tol_quantics=1e-8, maxbonddim_quantics=50)</code>	Aubry–André–Harper: $H = -t \sum c_{i+1}^\dagger c_i + V \cos(2\pi b i + \phi) n_i$.

HUniform2DSquare(Lx, Ly, t; tol_quantics=1e-8, maxbondddim_quantics=10, cutoff=1e-10)	Uniform square lattice tight-binding: <code>kineticintra2DNNN</code> ($nn=1$) + <code>kineticNNN</code> ($nn=N_x$).
HUniform2Dhex(Lx, Ly, t; tol_quantics=1e-8, maxbondddim_quantics=10, cutoff=1e-10)	Uniform honeycomb lattice: <code>kineticintra2DNNhex</code> (checkerboard intra-row mask) + <code>kineticNNN</code> ($nn=N_x$).
HUniform2Dtri(Lx, Ly, t; tol_quantics=1e-8, maxbondddim_quantics=10, cutoff=1e-10)	Uniform triangular lattice (three kinetic terms: intra-row NN, y -hop, and both triangular diagonals via <code>kineticinterNNNtriSWNE/kineticinterNNNtriSEnw</code>).
HUniform2Dtri.bravais(Lx, Ly, t; tol_quantics=1e-8, maxbondddim_quantics=10, cutoff=1e-10)	Bravais triangular lattice: x , y , and diagonal bonds matching $\mathbf{a}_1 = (1, 0)$, $\mathbf{a}_2 = (\frac{1}{2}, \frac{\sqrt{3}}{2})$.
HChern8(Lx, Ly, V, t; a=5/64*2^Lx, t2=0.2t, tol_quantics=1e-8, maxbondddim_quantics=10, cutoff=1e-10)	8-fold Chern mosaic: alternating intra-row hopping plus complex NNN diagonal terms modulated by a 4-rotated-wavevector 8-fold pattern at amplitude V .
H2DChernhex(Lx, Ly, t, t2, ms; uniformhaldane=false, uniformsemenoff=false, tol_quantics=1e-8, maxbondddim_quantics=10, cutoff=1e-10)	Haldane Chern insulator on honeycomb; domain walls in t_2 and Semenoff mass at $x=N_x/2$ unless <code>uniformhaldane/uniformsemenoff</code> are set.
HQC2DSquare(Lx, Ly, t=1.0; tol_quantics=1e-8, maxbondddim_quantics=100, cutoff=1e-10)	Quasicrystal-modulated square lattice: bond-midpoint hopping amplitude from an 8-fold pattern with two competing wavevectors $b_1 = 5\sqrt{5}/2$, $b_2 = \sqrt{3}N_x/16$.
build_hamiltonian(model, L; mparams, mparam_dict)	Dispatch via <code>MODEL_REGISTRY</code> : build 1D MPO from model name and parameter string.
build_hamiltonian(model, Lx, Ly; mparams, mparam_dict)	2D variant.

Sublattice-based builders (return `TBHamiltonian` directly):

Function	Description
<code>honeycomb_sublattice_hamiltonian(Lx, Ly, t=1; cutoff=1e-8, maxdim=200) → TBH</code>	Honeycomb NN Hamiltonian with 2-atom unit cell. 3 bond types: intra-cell A-B, x-intercell B→A, y-intercell B→A. <code>sublattice_s</code> dim-2 postpended; <code>scale=3.5 t </code> .
<code>honeycomb_nnn_hamiltonian(Lx, Ly, t=1, t2=0; cutoff=1e-8, maxdim=200) → TBH</code>	Honeycomb NN+NNN: adds same-sublattice NNN hopping t_2 (identity in sublattice space, $A \rightarrow A$ and $B \rightarrow B$) along the three triangular Bravais directions on top of NN t . <code>scale=3.5 t + 3.5 t2 </code> .

<code>kagome_hamiltonian(Lx, Ly, t=1; t_AB=t, t_AC=t, t_BC=t, cutoff=1e-8, maxdim=200) → TBH</code>	Kagomé lattice, 3-atom unit cell (A, B, C). Each of <code>t_AB</code> (intra-cell + x shift), <code>t_AC</code> (intra-cell + y shift), <code>t_BC</code> (intra-cell + diagonal shift N_x-1) defaults to <code>t</code> ; pass individually for anisotropic/breathing kagomé. Flat band at $E = -t$ (uniform case). <code>sublattice_s</code> dim-3 postpended; <code>scale=4.5 max(t_AB , t_AC , t_BC)</code> .
<code>lieb_hamiltonian(Lx, Ly, t=1; t_AB=t, t_AC=t, cutoff=1e-8, maxdim=200) → TBH</code>	Lieb lattice, 3-atom unit cell (A corner, B x -edge, C y -edge); no B-C bond. <code>t_AB</code> (A→B, intra-cell + x shift) and <code>t_AC</code> (A→C, intra-cell + y shift) default to <code>t</code> . Flat band at $E = 0$. <code>sublattice_s</code> dim-3 postpended; <code>scale=2.5 max(t_AB , t_AC)</code> .
<code>dice_hamiltonian(Lx, Ly, t=1; t_AB=t, t_AC=t, cutoff=1e-8, maxdim=200) → TBH</code>	Dice (T3) lattice, 3-atom unit cell: hub site A (coordination 6) couples to rim sites B and C. <code>t_AB</code> controls A-B bonds (intra-cell, x, y); <code>t_AC</code> controls A-C bonds (no intra-cell bond; x, y , and diagonal shift N_x+1). Doubly-degenerate flat band at $E = 0$. <code>sublattice_s</code> dim-3 postpended; <code>scale=4.5 max(t_AB , t_AC)</code> .
<code>ssh_sublattice_hamiltonian(L, t=1, d=0; cutoff=1e-8, maxdim=200) → TBH</code>	SSH chain with an explicit 2-component sublattice index (A=1, B=2, postpended). Intra-cell A→B amplitude $t + d$; inter-cell $B(n) \rightarrow A(n+1)$ amplitude $t - d$ via <code>generate_kin_u/generate_kin_d</code> . Periodic (the binary-increment wrap-around gives the $B(N-1) \rightarrow A(0)$ bond). Unlike the other sublattice builders this constructor also sets <code>geometry/geometry_uc</code> : atom i in unit cell $n = \lfloor (i-1)/2 \rfloor$ sits at $[n]$ (A) or $[n+0.5]$ (B), with <code>geometry_uc(i) = [n]</code> for both. <code>scale=1.1(t+d + t-d)</code> .

Known models in `MODEL_REGISTRY`: "uniform", "ssh", "aah", "square_2d", "hex_2d", "triangular_2d", "triangular_bravais", "chern8", "chernhex", "qc2dsquare". Sublattice-aware geometries ("honeycomb", "honeycomb_nnn", "kagome", "lieb", "dice", "ssh_sublattice") are routed directly by `get_Hamiltonian` and are not in the plain `MODEL_REGISTRY`. `_parse_param_string(s) → Dict{Symbol,Any}` is the internal helper behind the `mparams` string: it splits "key1=val1, key2=val2" on commas/whitespace and auto-types each value as `Bool`, `Int`, `Float64`, or `String`.

Generic n th-Neighbor Hopping (nnneighbor_tk.jl)

These functions operate on any 2D `TBHamiltonian` regardless of geometry, adding n th-nearest-neighbor bonds detected automatically via a reference patch.

Shell Detection

Signature	Description
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<code>get_shell_disps(H, nn; Lx, Ly) → Vector{NTuple{4,Int}}</code>	Print and return the canonical displacement tuples $(\Delta i_x, \Delta i_y, s_{\text{from}}, s_{\text{to}})$ for the n th-nearest-neighbour shell. “Canonical” means $\Delta i_x > 0$, or $(\Delta i_x=0 \text{ and } \Delta i_y > 0)$, or $(\Delta i_x=\Delta i_y=0 \text{ and } s_{\text{from}} < s_{\text{to}})$. Call this first to discover which tuples will be passed to an amplitude function.
---	--

Hopping Accumulator

Signature	Description
<code>add_hopping_2D!(H, f; Lx, Ly, nn=1, layer=nothing, lattice=nothing, geometry=nothing, maxdim=50, tol=1e-8) → H</code>	Add nth-nearest-neighbor hopping to H in place. Iterates over canonical shell displacements; for each, builds a forward+backward shift MPO weighted by the amplitude returned by f. Works for all 2D geometries including sublattice-aware ones (honeycomb, kagomé, Lieb, dice). Layered Hamiltonians ($H.\text{layer_s}$ set, requires $H.\text{spin_s}/H.\text{nambu_s}$ both <code>nothing</code>): <code>layer=nothing</code> applies the hopping to every layer; an integer or collection of integers (1-indexed) restricts it to those layers. The shell term is built once on a scratch position(+sublattice) Hamiltonian and prepended per selected layer via <code>prepend_layer_projector</code>. If $H.\text{geometry}$ is not already set, pass <code>lattice=:square/:triangular/:honeycomb</code> or an explicit <code>geometry</code> (a function $i \rightarrow r_i$, or an $N \times d$ position matrix with one row per position site, or per atom when $H.\text{sublattice_s}$ is set).

Amplitude function forms accepted by f :

Signature of f	Meaning
<code>f::Number</code>	Uniform amplitude for all bonds in the shell.
<code>f($\Delta i_x, \Delta i_y, fs, ts$) → Number</code>	Direction-dependent; receives canonical displacement and sublattice indices.
<code>f(n) → Number</code>	Site-dependent (0-indexed unit-cell index n).
<code>f(ix, iy) → Number</code>	Site-dependent via explicit 2D unit-cell coordinates.
<code>f($n, \Delta i_x, \Delta i_y, fs, ts$) → Number</code>	Both site (0-indexed) and direction.
<code>f($ix, iy, \Delta i_x, \Delta i_y, fs, ts$) → Number</code>	Full control: explicit 2D coords and direction.

Sublattice convention: For sublattice-aware geometries, fs/ts are 1-indexed sublattice numbers. The forward operator uses matrix element $M_{\text{fwd}}[s_{\text{from}}, s_{\text{to}}] = 1$, consistent with the original Hamiltonian builders.

Typical usage:

```
# Inspect which displacements will be passed (honeycomb, nn=2)
TensorBinding.get_shell_disps(H, 2; Lx=Lx, Ly=Ly)
# [(1,0,1,1), (0,1,1,1), (1,-1,1,1), (1,0,2,2), (0,1,2,2), (1,-1,2,2)]

# Uniform NNN hopping (same-sublattice, real amplitude)
add_hopping_2D!(H, 0.2; Lx=Lx, Ly=Ly, nn=2)

# Direction-dependent complex amplitude (Haldane-like phases)
add_hopping_2D!(H,
  (dx,dy,fs,ts) -> (dx,dy,fs,ts)==(1,0,1,1) ? t2*exp(im*phi) : 0.0im;
  Lx=Lx, Ly=Ly, nn=2)

# Position+direction dependent modulated hopping
add_hopping_2D!(H,
  (ix,iy,dx,dy,fs,ts) -> t1 + amp*sin(dot(G_vec[(dx,dy,fs,ts)], r_mid(ix,iy)));
  Lx=Lx, Ly=Ly, nn=1)
```

Bilayer and Multilayer Hamiltonians**Commensurate Stacking (bilayer_tk.jl)**

Site encoding: site 1 = layer qubit (Index of dim n_{layers}); sites $2 \dots L + 1$ = position qubits.
Constructors return a TBHamiltonian with H.layer_s set.

Signature	Description
<code>bilayer_hamiltonian(lattice, Lx, Ly; stacking=:AA, t_intra=1.0, t_inter=0.3, sublattice=false, cutoff=1e-8, maxdim=200) → TBHamiltonian</code>	Two-layer Hamiltonian: $\sum_k P_k \otimes H_{\text{mono}} + (1\rangle\langle 2 + 2\rangle\langle 1) \otimes V$. <code>stacking=:AA</code> (on-site) or <code>:Bernal</code> (A_1–B_2, honeycomb only). <code>sublattice=true</code> appends an explicit sublattice Index (<code>H.sublattice_s</code>, <code>dim=2</code>, currently <code>:honeycomb</code> only): the monolayer term comes from <code>honeycomb_sublattice_hamiltonian</code> and interlayer coupling is built per-sublattice (on-site identity for <code>:AA</code>, or $A_1 \rightarrow B_2/B_2 \rightarrow A_1$ for <code>:Bernal</code>), with <code>H.aux_side=:pre</code>.
<code>multilayer_hamiltonian(lattice, Lx, Ly, n_layers; stacking=:AA, t_intra=1.0, t_inter=0.3, sublattice=false, cutoff=1e-8, maxdim=200) → TBHamiltonian</code>	Generalises to n_{layers} layers; interlayer coupling between adjacent pairs only. <code>sublattice=true</code> behaves as for <code>bilayer_hamiltonian</code>, applied to every adjacent layer pair $(k, k+1)$.
<code>interlayer_mpo(lattice, stacking, Lx, Ly, sites; t_inter=1.0, cutoff=1e-8) → MPO</code>	Build the position-space interlayer coupling MPO V. Routes to <code>_bernal_interlayer_mpo</code> or <code>_aa_interlayer_mpo</code>.

<code>_bernal_interlayer_mpo(L, sites; t_inter=1.0, cutoff=1e-8) → MPO</code>	Exact Bernal A ₁ -B ₂ coupling: $V = t_{\perp}(K_u D_A + D_A K_d)$. D_A projects onto odd (A) sites (1-indexed); $K_{u/d}$ are ± 1 shifts.
<code>_aa_interlayer_mpo(sites; t_inter=1.0) → MPO</code>	AA stacking: on-site coupling $V = t_{\perp} \mathbf{I}$.

Twisted Multilayer (twisted.tk.jl)

Signature	Description
<code>twisted_multilayer_hamiltonian(lattice, Lx, Ly, angles_deg; t_intra=1.0, t_inter=0.3, $\alpha_{\text{decay}}=1/16$, tol=1e-6, cutoff=1e-8, maxdim=200) → TBHamiltonian</code>	Twisted multilayer: layer k rotated to <code>angles_deg[k]</code> . Interlayer coupling $V_{kl}(i, j) = t_{\perp} e^{-\alpha \mathbf{r}_k^{(i)} - \mathbf{r}_l^{(j)} }$ compressed via QTCI.
<code>twisted_bilayer_hamiltonian(lattice, Lx, Ly, θ_{deg}; ...) → TBHamiltonian</code>	Convenience wrapper: layer 1 at 0° , layer 2 at θ_{deg} .
<code>monolayer_hamiltonian(lattice, Lx, Ly, sites; t=1.0, cutoff=1e-8) → MPO</code>	Single-layer NN Hamiltonian MPO for <code>:square</code> , <code>:triangular</code> , or <code>:honeycomb</code> .

Layer-Core Prepend/Postpend Helpers

These functions extend an L -site MPO by one site by prepending or postpending a single-site layer operator. All are thin wrappers around `prepend_op/postpend_op`.

Signature	Description
<code>prepend_layer_projector(H, layer_s, k) → MPO</code>	Prepend $ k\rangle\langle k $ (1-indexed). Used for $P_k \otimes H$.
<code>prepend_layer_hopping(H, layer_s, k, l) → MPO</code>	Prepend $ k\rangle\langle l $. Used for interlayer $ k\rangle\langle l \otimes V$.
<code>postpend_layer_projector(H, layer_s, k) → MPO</code>	Postpend $ k\rangle\langle k $ to the end of H.
<code>postpend_layer_hopping(H, layer_s, k, l) → MPO</code>	Postpend $ k\rangle\langle l $ to the end of H.

Spin and Nambu Extensions (Supercond.tk.jl)

Index Constructors

Function	Description
<code>spin_index()</code>	Create a dim-2 Index tagged "Spin" (1= $\uparrow$, 2= $\downarrow$).
<code>nambu_index()</code>	Create a dim-2 Index tagged "Nambu" (1=particle, 2=hole).

Unified Prepend/Postpend Operators (utils.jl)

Signature	Description
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<code>prepend_op(H, s, mat) → MPO</code>	General prepend: extends H by prepending an arbitrary matrix <code>mat</code> on site <code>s</code> .
<code>prepend_op(H, s, op::Symbol) → MPO</code>	Named operator lookup for "Spin" or "Nambu" tagged indices.
<code>prepend_op(H, s, k, l) → MPO</code>	Sparse $ k\rangle\langle l $ on site <code>s</code> (any dimension; 1-indexed).
<code>prepend_op(H, s, k) → MPO</code>	Projector $ k\rangle\langle k $.
<code>postpend_op(H, s, mat / op / k, l / k) → MPO</code>	Symmetric counterpart of <code>prepend_op</code> ; appends to the end of H .

Spin and Nambu Wrappers

Signature	Description
<code>prepend_spin(H, spin_s, op) → MPO</code>	Prepend a spin- $\frac{1}{2}$ operator. <code>op</code> is a <code>Symbol</code> or 2×2 matrix. Built-in ops: <code>:Id</code> , <code>:Pup</code> , <code>:Pdn</code> , <code>:Sz</code> , <code>:Sp</code> , <code>:Sm</code> , <code>:Sx</code> , <code>:Sy</code> , <code>:iSy</code> , <code>:miSy</code> .
<code>postpend_spin(H, spin_s, op) → MPO</code>	Postpend a spin- $\frac{1}{2}$ operator to the end of H . Same ops as above.
<code>prepend_nambu(H, nambu_s, op) → MPO</code>	Prepend a Nambu (particle-hole) operator. Built-in ops: <code>:Id</code> , <code>:Pp</code> , <code>:Ph</code> , <code>:tz</code> , <code>:tx</code> , <code>:ty</code> , <code>:tp</code> , <code>:tm</code> .
<code>postpend_nambu(H, nambu_s, op) → MPO</code>	Postpend a Nambu operator to the end of H . Same ops as above.

Antisymmetric Pairing MPO Builders

Pairing terms $\Delta(i, j)$ for `bdg_hamiltonian/bdg_spin_hamiltonian` must be antisymmetric, $\Delta(i, j) = -\Delta(j, i)$, since $c_i^\dagger c_j^\dagger = -c_j^\dagger c_i^\dagger$. These builders construct such Δ as an MPO `H_pair`.

Signature	Description
<code>pairingNNN(L, sites, hopping, nn; apply_kwargs=NamedTuple()) → MPO</code>	Exact n th-neighbor antisymmetric pairing on a 1D chain: $H_{\text{pair}} = \text{hopping} \cdot K_f^{nn} - K_b^{nn} \cdot \text{dag}(\text{hopping})$, where $K_{f/b}$ are the quantics forward/backward unit-shift MPOs (built via <code>compose_power</code>). Antisymmetric analogue of <code>kineticNNN</code> . For uniform pairing $\text{hopping} = \Delta \cdot I$ this gives $H_{\text{pair}} = \Delta \cdot (K_f^{nn} - K_b^{nn})$.
<code>pairing2MPO(f, N, sites; tol=1e-8, type=ComplexF64, initial_positions=[], unfoldingscheme=:interleaved) → MPO</code>	QTCI compression of a general pairing matrix $\Delta(i, j) = f(i, j)$; thin wrapper around <code>hopping2MPO</code> . The caller must supply an antisymmetric <code>f</code> ($f(i, j) = -f(j, i)$). Prefer <code>pairingNNN</code> for nearest-neighbor pairing (exact, no TCI); use <code>pairing2MPO</code> for longer-range or d -wave/ $p \pm ip$ patterns.

Higher-Level Assemblers

Signature	Description
-----------	-------------

<code>spin_hamiltonian(H_up, H_dn, spin_s; H_Zeeman=nothing, cutoff=1e-8) → MPO</code>	$P_{\uparrow} \otimes H_{\uparrow} + P_{\downarrow} \otimes H_{\downarrow} [+S_z \otimes H_{\text{Zeeman}}]$.
<code>bdg_hamiltonian(H_kin, H_pair, nambu_s; cutoff=1e-8) → MPO</code>	Spinless BdG: $\tau_z \otimes H_{\text{kin}} + \tau_+ \otimes H_{\text{pair}} + \tau_- \otimes H_{\text{pair}}^{\dagger}$.
<code>bdg_spin_hamiltonian(H_up, H_dn, H_pair, spin_s, nambu_s; H_soc=nothing, cutoff=1e-8) → MPO</code>	Spin- $\frac{1}{2}$ singlet BdG in Nambu–spin space. Pairing: $\tau_+ \otimes (i\sigma_y) \otimes H_{\text{pair}} + \text{h.c.}$. Optional Ising SOC $\tau_z \otimes S_z \otimes H_{\text{soc}}$.

Kernel Polynomial Method (KPM.tk.jl)

The KPM expands spectral quantities in Chebyshev polynomials $T_n(\tilde{H})$ where $\tilde{H} = (H - \text{center})/\text{scale} \in [-1, 1]$. Supported kernels: `:jackson` (default), `:lorentz`, `:fejer`, `:dirichlet`, `:hodc`.

Three complementary Chebyshev strategies are available, trading memory for generality:

- **MPO mode** (`mode=:mpo`): stores the full operator $T_n(\tilde{H})$ as an MPO. Needed for spectral functions at arbitrary sites, Green’s functions, and density matrices. Bond dimension grows with n ; control with `maxdim`.
- **Diagonal/online mode**: extracts only the diagonal of each T_n as an MPS. Scales as $O(N_{\text{cheb}})$ MPO operations regardless of the number of energy points. Used by `get_ldos_spectrum`, `get_ldos_online`, and `get_ldos_spatial`.
- **MPS mode** (`mode=:mps`): propagates a reference state $|\phi_n\rangle = T_n(\tilde{H})|\psi_0\rangle$. Site-resolved LDoS at ψ_0 without storing any operator. Required for exciton systems (2L-site chain makes MPO mode prohibitive).

Spectral Bounds Estimation

Signature	Description
<code>_estimate_spectral_bounds(H_mpo, sites; dmrg_nsweeps=5, dmrg_maxdim=[10,20,40], dmrg_linkdim=4) → (scale, center)</code> <code>_ensure_scale!(H::TBHamiltonian; ...)</code>	Run DMRG to minimise H and $-H$, yielding $E_{\min}/E_{\max}$. Returns <code>scale = (E_{max} - E_{min})/2 × 1.1</code> and <code>center = (E_{max} + E_{min})/2</code> . If <code>H.scale==0</code> , call <code>_estimate_spectral_bounds</code> and store in <code>H.scale/H.center</code> . No-op otherwise. Called lazily by all high-level KPM functions.

Building the Chebyshev List

Signature	Description
<code>KPM_Tn(H_mpo, N, sites; scale=nothing, center=0.0, maxdim=40, cutoff=1e-8, ...) → (Tn_list, scale, center)</code>	Low-level MPO form. Build Chebyshev MPOs $T_n(\tilde{H})$ for $n = 0 \dots N$. If <code>scale=nothing</code> , bounds estimated via DMRG.

<code>KPM_Tn(H::TBHamiltonian, Ncheb; mode=:mpo, psi0=nothing, maxdim=40, cutoff=1e-8, ...) → (Tn_list, scale, center)</code>	High-level form. Calls <code>_ensure_scale!</code> lazily. <code>mode=:mpo</code> : caches MPO list in <code>H._tn_cache</code> . <code>mode=:mps</code> : requires <code>psi0</code> ; caches MPS list in <code>H._tn_mps_cache</code> . Both update <code>H._tn_Ncheb</code> .
<code>KPM_Tn(H::TBHamiltonian, Ncheb, X; maxdim=40, cutoff=1e-8, ...) → (Tn_mps_list, scale, center)</code>	Exciton convenience form. Builds MPS chain $T_n(\tilde{H}) X, X\rangle$ for 1-indexed exciton site X ; caches in <code>H._tn_mps_cache</code> . No MPO mode provided for 2L-site exciton systems.
<code>KPM_Tn_mps(H_mpo, N, psi0, sites; scale=nothing, center=0.0, maxdim=40, cutoff=1e-8, ...) → (Tn_mps_list, scale, center)</code>	Low-level MPS form. Builds $ \phi_n\rangle = T_n(\tilde{H}) \psi_0\rangle$ for $n = 0 \dots N$. Moments: $\mu_n = \langle \psi_0 \phi_n \rangle$.
<code>KPM_Tn_mps(H::TBHamiltonian, N, psi0; maxdim=40, cutoff=1e-8, ...) → (Tn_mps_list, scale, center)</code>	High-level MPS form. Calls <code>_ensure_scale!</code> ; caches in <code>H._tn_mps_cache</code> .

Important — return order: The return tuple is `(Tn_list, scale, center)`. A common bug is `Tn, _, scale = KPM_Tn(...)`, which assigns `center` to `scale`. The correct pattern is:

```
Tn, scale, center = KPM_Tn(H, Ncheb, sites)
omega_r = (omega - center) / scale    # rescaled energy in (-1, 1)
```

When using the `TBHamiltonian` overload, `H.scale` and `H.center` are updated automatically and can be read directly.

Spectral Observables from Stored Chebyshev List

Signature	Description
<code>get_ldos_w_from_Tn(Tn_list, N, ω; maxdim=40, kernel=:jackson, ...) → MPO</code>	Full spectral MPO $\delta(\omega - H)$ at rescaled $\omega \in (-1, 1)$. Legacy full-MPO path; prefer <code>get_ldos</code> with <code>mode=:diag</code> for site-resolved LDOS.
<code>get_ldos_diag_from_Tn(Tn_list, N, ω_vals; kernel=:jackson, lambda=4.0, maxdim=40, cutoff=1e-8) → Vector{Union{Nothing,MPS}}</code>	Diagonal-MPS LDOS at all energies in a single pass over <code>Tn_list</code> . Returns one MPS $A(r, \omega)$ per energy (<code>nothing</code> for $ \omega \geq 1$).
<code>get_Green_retarded_from_Tn(Tn_list, N, ω; $\eta=1e-2$, maxdim=40, kernel=:jackson, ...) → MPO</code>	Retarded Green's function MPO $G^R(\omega + i\eta) = (\omega + i\eta - H)^{-1}$.
<code>get_density_from_Tn(Tn_list, N; fermi=0.0, maxdim=40, kernel=:jackson, ...) → MPO</code>	Density matrix $\rho = \theta(\mu - H)$ at rescaled Fermi level <code>fermi</code> .
<code>get_ldos_from_mun(mun_list, N, E; kernel=:jackson, lambda=4.0, ...) → Real</code>	Reconstruct LDoS at rescaled energy E from moment list $\mu_n = \langle \psi_0 T_n(\tilde{H}) \psi_0 \rangle$ (MPS mode output).
<code>get_ldos_hodc_from_mun(mun_list, N, E; eta=0.02, m_order=4) → Real</code>	HODC (Higher-Order Delta Chebyshev) variant of <code>get_ldos_from_mun</code> ; contour-based kernel giving sharper spectral features than Jackson, at the cost of <code>m_order</code> extra parameters.

High-Level LDOS Dispatch (TBHamiltonian interface)

Signature	Description
<code>get_ldos(H, ω_{phys}; mode=:diag, psi0=nothing, kernel=:jackson, lambda=4.0, eta=0.0, m_order=4, maxdim=40, cutoff=1e-8, z1, w1) $\rightarrow$ MPS or MPO or Real</code>	Unified LDOS dispatch. Requires a prior <code>KPM.Tn</code> call to populate the cache. <code>mode=:diag</code> (default): returns diagonal MPS (requires MPO cache). <code>mode=:mpo</code> : returns full spectral MPO (requires MPO cache; legacy). <code>mode=:mps</code> : returns <code>Real</code> from moments (requires MPS cache + <code>psi0</code>).
<code>get_ldos_spectrum(H, $\omega_{\text{phys_vals}}$; kernel=:jackson, lambda=4.0, maxdim=40, cutoff=1e-8) $\rightarrow$ Vector{Union{Nothing,MPS}}</code>	Site-resolved LDOS at all energies in a single pass over the cached MPO list. Real-space analogue of <code>get_bands</code> . Requires MPO cache.
<code>get_ldos_online(H, Ncheb, X, $\omega_{\text{phys_vals}}$; kernel=:jackson, lambda=4.0, maxdim=100, cutoff=1e-8, verbose=false) $\rightarrow$ Vector{Float64}</code>	Online LDoS at 1-indexed site X for all energies, using only 3 MPS simultaneously. Never stores any Chebyshev cache. Works for standard and exciton Hamiltonians (exciton: evaluation state is $ X, X\rangle$).

```

get_ldos_spatial(H::TBHamiltonian,
Ncheb,  $\omega_{\text{phys\_vals}}$ ; num_x=H.N,
num_y=nothing, num_avg=1, mode=:mpo,
x_start=1, x_end=H.N,
x_groups=nothing, grid=false,
xwin=nothing, ywin=nothing,
box_half=0, reduce=:point,
sublattice=:auto, kernel=:jackson,
lambda=4.0, maxdim=100, cutoff=1e-8,
verbose=false, nambu_proj=false,
proj_nambu=nothing, spin_proj=false,
proj_s=nothing, layer_proj=false,
proj_layer=nothing,
sublat_proj=false, proj_sl=nothing) →
Matrix{Float64}

```

Spatially-resolved LDOS, real-space analogue of `get_bands`; returns an $N_\omega \times N_x$ matrix. Sample positions/blocks come from `spatial_sampling_plan` (built from `H.L`, `H.Lx`, and the kwargs below).

`reduce=:point` (default): read the LDOS at `num_x[×num_y]` sample cells (`box_half>0` averages a $(2 \text{ box_half} + 1)^2$ box); cheap but can alias features narrower than the grid.

`reduce=:block` (`mode=:mpo` only): partition into `num_x×num_y` power-of-two blocks and report the block integral via partial contraction — gap-free, catches thin in-gap/edge states. `grid=true` lays centers on a 2D `num_x×num_y` unit-cell grid; `xwin/ywin` give 0-indexed unit-cell window bounds for zooming into a patch.

`sublattice` controls the column layout when `H.sublattice_s` is set: `:auto` (default) resolves per-atom columns ($N_\omega \times (n_g n_{\text{sub}})$, atom order $[A_0, B_0, A_1, B_1, \dots]$) at full unit-cell resolution and averages sublattices into one column per cell ($N_\omega \times n_g$) for coarser grids or `box_half>0`; `:resolve/:average` force one or the other, and `proj_sl=k` always resolves sublattice k alone. `mode=:mpo` (default): single Chebyshev pass, cost independent of `num_x/n_sub`. `mode=:mps`: independent MPS recursion per (position, sector). `nambu_proj/proj_nambu`, `spin_proj/proj_s`, `layer_proj/proj_layer`, `sublat_proj/proj_sl`: same auxiliary-DOF projection interface as `get_bands`; auto-enabled when the corresponding aux index is present.


```
get_dos_stochastic(H::TBHamiltonian,
Ncheb,  $\omega_{\text{phys}}$ _vals; N_sample=50,
N_bound=0, seed=42, normalize=false,
dos_weighting=:trace,
kernel=:jackson, lambda=4.0, eta=0.0,
m_order=4, maxdim=100, cutoff=1e-8,
verbose=false, nambu_proj=false,
proj_nambu=nothing, spin_proj=false,
proj_s=nothing, layer_proj=false,
proj_layer=nothing,
sublat_proj=false, proj_sl=nothing) →
Vector{Float64}
```

Stochastic full DoS via random trace estimation (MPS Chebyshev, 3 MPS per sample). `dos_weighting=:trace` (default) returns the trace DoS $\text{Tr}[\delta(\omega - H)]$, divided by the traced dimension if `normalize=true`; `dos_weighting=:sample` returns the raw sample-averaged signal (useful to inspect an exciton bound peak before sector-size weighting), and `normalize` is then ignored. Reconstruction uses `_dos_weight_matrix`: convolution kernels (`:jackson` default, `:lorentz` via `lambda`, `:fejer`, `:dirichlet`) or `kernel=:hodc` (`eta`, `m_order`; `eta=0` → $1/(N_{\text{cheb}}+1)$). Unlike `get_ldos_spatial`, auxiliary-DOF projections (`nambu_proj`, `layer_proj`, `sublat_proj`, ...) are *not* auto-enabled — the default samples the full D -dimensional Hilbert space; setting any `*_proj=true` samples position states with fixed aux sectors ($N_{\text{phys}} \times n_{\text{sectors}}$ effective states). For exciton Hamiltonians (`length(H.sites)==2H.L`) with no aux projections, `N_bound>0` dedicates samples to the bound sector $|x, x\rangle$, combined as $(D - N_{\text{phys}})\bar{A}_{\text{full}} + N_{\text{phys}}\bar{A}_{\text{bound}}$; `N_bound=0` (default) is uniform sampling over all D states.

Spatial Exciton LDOS

Signature	Description
<pre>get_exciton_ldos_spatial(H::TBHamiltonian, Ncheb, ω_{phys}_vals; X_list=nothing, X_groups=nothing, x_groups=nothing, num_x=H.N, num_avg=1, x_start=1, x_end=H.N, kernel=:jackson, lambda=4.0, eta=0.0, m_order=4, maxdim=100, cutoff=1e-8, verbose=false, printinfo=false) → Matrix{Float64}</pre>	Spatially-resolved exciton LDOS. For each bound exciton position X (electron = hole = X, 1-indexed in $1:H.N$), runs an online MPS Chebyshev recursion from $X, X\rangle$ and accumulates all requested energies in one pass; no Chebyshev cache is stored on H. Rows are energies, columns are positions/groups. <code>X_list</code> selects positions directly; <code>X_groups</code> (alias <code>x_groups</code>) averages several bound-pair probes into one output column. If neither is given, <code>num_x</code> coarse groups are generated over <code>x_start:x_end</code> with <code>num_avg</code> sub-positions per group. <code>kernel=:hodc</code> uses the HODC reconstruction (<code>eta</code>, <code>m_order</code>); otherwise <code>:jackson/:lorentz/:fejer/:dirichlet</code>.

<pre> get_exciton_ldos(H::TBHamiltonian, X::Int, omega_phys::Real; Ncheb=200, kernel=:jackson, lambda=4.0, eta=0.0, m_order=4, maxdim=40, cutoff=1e-8, verbose=false) → Real get_exciton_ldos(H::TBHamiltonian, X::Int, omega_phys_vals; ...) → Vector{Float64} get_exciton_ldos_separation(H::TBHamiltonian, Ncheb, omega_phys_vals; d_list, R_list=1:H.N, boundary=:open, kernel=:jackson, lambda=4.0, eta=0.0, m_order=4, maxdim=100, cutoff=1e-8, verbose=false, printinfo=false) → Array{Float64,3} exciton_radius2(rho::AbstractArray{<:Real}, d_list) → Matrix{Float64} </pre>	Single-position convenience wrappers around <code>get_exciton_ldos_spatial</code> with <code>X_list=[X]</code>: the scalar overload returns <code>ldos[1,1]</code> for one energy, the vector overload returns <code>vec(ldos[:,1])</code> for a list of energies. Relative-separation-resolved exciton LDOS $\rho(d, R, E) = \langle R+d, R \delta(E - H) R+d, R \rangle$, generalising <code>get_exciton_ldos_spatial</code> (the $d = 0$ slice) to electron-hole probes with electron at $R + d$ and hole at R (both 1-indexed in $1:H.N$). Each (d, R) pair is a single product-state probe <code>mpsexciton(R+d, R, H.sites)</code> — no carry/superposition construction is needed, unlike the momentum-space probes. <code>boundary=:open</code> (default) leaves out-of-range $R + d$ as <code>NaN</code>; <code>:periodic</code> wraps modulo $H.N$. Returns <code>(Nomega, length(d_list), length(R_list))</code>; cost is <code>length(d_list)*length(R_list)</code> Chebyshev recursions. Energy-resolved exciton radius $\xi^2(R, E) = \sum_d d^2 P(d R, E)$ with $P(d R, E) = \rho(d, R, E) / \sum_{d'} \rho(d', R, E)$, from a <code>rho</code> array returned by <code>get_exciton_ldos_separation</code>. Returns <code>(Nomega, n.R)</code>; entries are <code>NaN</code> where any d-slice of <code>rho</code> contains <code>NaN</code> or the normalisation $\sum_d \rho(d, R, E)$ vanishes.
---	--

`_get_exciton_ldos_cached` is a legacy cache-backed scalar helper (Chebyshev moments via `KPM.Tn`); the public exciton LDOS API above routes through `get_exciton_ldos_spatial` instead, which performs online MPS recursion without storing a cache on `H`.

HODC Kernel

Signature	Description
<code>compute_hodc_params(m=6) → (z1, w1)</code>	Compute m quadrature nodes <code>z1</code> and weights <code>w1</code> for the HODC kernel. Call once; pass to KPM functions as <code>kernel=:hodc, z1=z1, w1=w1</code> .

Density Matrix Purification (Purification.tk.jl)

Iterative methods that drive eigenvalues of an approximate density matrix to exactly 0 or 1, refining the zero-temperature projector $\theta(\mu - H)$ without Gibbs ringing.

Initial Guess

Signature	Description
<code>purification_initial_guess(H_mpo, scale, sites; maxdim=40, cutoff=1e-8) → MPO</code>	Low-level form. Build $\rho_0 = \frac{1}{2}I - \frac{1}{2s}H$. Works only for center-symmetric spectra.
<code>purification_initial_guess(H::TBHamiltonian, $\epsilon_F=0.0$, maxdim=40, cutoff=1e-8) → MPO</code>	High-level form. Calls <code>_ensure_scale!</code> automatically. Uses formula accounting for nonzero center and Fermi-level shift ϵ_F : $\rho_0 = \left(\frac{1}{2} + \frac{\epsilon_F + \text{center}}{2\text{scale}}\right)I - \frac{H}{2\text{scale}}.$

McWeeny Purification

Signature	Description
<code>mcweeny_purify(ρ_0; maxiters=30, maxdim=40, cutoff=1e-8, tol=1e-5, verbose=false) → MPO</code>	Iterate $\rho_{n+1} = 3\rho_n^2 - 2\rho_n^3$ until $\ \rho^2 - \rho\ /\ \rho\ < \text{tol}$. Quadratic convergence; 2 MPO products per step.
<code>mcweeny_purify(H::TBHamiltonian; $\epsilon_F=0.0$, maxiters=30, maxdim=40, cutoff=1e-8, tol=1e-5, verbose=false) → MPO</code>	High-level form. Builds initial guess from H (optionally shifted by ϵ_F), runs McWeeny, caches result in H._density_cache.

SP2 Purification

Signature	Description
<code>sp2_purify(ρ_0, Nel; maxiters=40, maxdim=40, cutoff=1e-8, tol=1e-5, verbose=false) → MPO</code>	Iterate: if $\text{Tr}(\rho^2) \geq N_{\text{el}}$: $\rho_{n+1} = \rho^2$; else $\rho_{n+1} = 2\rho - \rho^2$. Drives $\text{Tr}(\rho) \rightarrow N_{\text{el}}$; 1 MPO product per step.
<code>sp2_purify(H::TBHamiltonian; Nel=H.N÷2, maxiters=40, maxdim=40, cutoff=1e-8, tol=1e-5, verbose=false) → MPO</code>	High-level form. Nel defaults to half-filling. Caches result in H._density_cache.

Unified Density-Matrix Wrapper

Signature	Description
<code>get_density(H::TBHamiltonian; method=:mcweeny, $\epsilon_F=0.0$, Ncheb=150, kernel=:jackson, lambda=4.0, maxdim=40, cutoff=1e-8, Nel=H.N÷2, maxiters=30, tol=1e-5, verbose=false) → MPO</code>	Compute and cache the zero-temperature density matrix $P = \theta(\epsilon_F - H)$. Returns H._density_cache immediately if already populated (set it to <code>nothing</code> to force recomputation). <code>method</code> : <code>:mcweeny</code> (McWeeny purification, default), <code>:sp2</code> (SP2 purification, requires Nel), or <code>:kpm</code> (KPM Chebyshev expansion of the Fermi step function, reusing/building H._tn_cache at order Ncheb; only <code>:jackson</code> kernel is meaningful for the step function).

Sign of an MPO

Signature	Description
<code>sign_mpo(A::MPO, sites; scale=1.0, maxdim=500, cutoff=1e-8, maxiters=30, tol=1e-5, verbose=false) → MPO</code>	Compute $\text{sign}(A) = \theta(A) - \theta(-A)$ for a Hermitian MPO A via two McWeeny purifications of $\rho_{0\pm} = (I \pm A/\text{scale})/2$. <code>scale</code> must bound the spectral radius of A (default 1.0, correct for operators with spectrum in $(-1, +1)$ such as <code>get_valley_operator</code>). Result has eigenvalues in $\{-1, +1\}$.

LDoS and DoS via Density-Matrix Derivative

Signature	Description
<code>get_ldos_drho(H::TBHamiltonian, ω; mode=:mpo, dmu=0.05, maxdim=40, cutoff=1e-8, maxiters=30, tol=1e-5, verbose=false) → MPO or MPS</code>	Local density of states $A(\omega) \approx [\rho(\omega + \delta\mu) - \rho(\omega - \delta\mu)]/(2\delta\mu)$ via finite-difference McWeeny density matrices, $\rho(\mu) = \theta(\mu - H)$. <code>mode=:mpo</code> (default) returns the full MPO ($\text{LDoS}(r, \omega) = \langle r A(\omega) r \rangle$); <code>mode=:mps</code> returns only the diagonal as an MPS (cheaper). A vector overload <code>get_ldos_drho(H, ωs::AbstractVector; ...)</code> applies this elementwise.
<code>get_dos_drho(H::TBHamiltonian, ω; dmu=0.05, maxdim=40, cutoff=1e-8, maxiters=30, tol=1e-5, verbose=false) → Float64</code>	Total density of states $\text{DOS}(\omega) \approx (\text{Tr}[\rho(\omega + \delta\mu)] - \text{Tr}[\rho(\omega - \delta\mu)])/(2\delta\mu)$. Cheapest variant: only scalar traces are computed. A vector overload <code>get_dos_drho(H, ωs::AbstractVector; ...)</code> → <code>Vector{Float64}</code> applies this elementwise.

Quantum Fourier Transform and Band Structure (QFT_tk.jl)

QFT Conjugation

Signature	Description
<code>conjugate_by_qft(W; tol=1e-9, maxdim=100) → MPO</code>	Returns $UWU^\dagger$ where U is the normalised QFT MPO (LSB at site 1). Maps any real-space MPO to momentum space.
<code>conjugate_by_qft(H::TBHamiltonian, W::MPO; tol=1e-9, maxdim=100) → MPO</code>	TBHamiltonian-aware overload. Applies $UWU^\dagger$ where U acts only on the position (Qubit) sites of H , with identity at every auxiliary site (layer, spin, sublattice, Nambu). Use when W lives on the full $H.\text{sites}$ space, e.g. after <code>replace_sites</code> .

`conjugate_by_qft_exciton(H::TBHamiltonian, particle (electron-hole) analogue for the W; tol=1e-9, maxdim=100) → MPO` interleaved 2L-site exciton encoding (`exciton_hamiltonian`). Applies $UWU^\dagger$ with $U = U_e \otimes U_h$ — two independent QFTs, one on the electron register (odd sites) and one on the hole register (even sites), $R = H.L$ qubits each. Sites are taken from `_pos_sites(H)`. Rotates an exciton spectral MPO T_n into (k_e, k_h) momentum space for band structure; pair with an *MPS-mode* KPM, since the MPO-mode Chebyshev list explodes in bond dimension for excitons.

Exciton Momentum-Space Spectra (MPS-KPM)

Momentum-resolved exciton spectra use the MPS mode end to end: conjugate the exciton Hamiltonian with `conjugate_by_qft_exciton`, build a fixed-total-momentum probe state, and run an online MPS Chebyshev recursion against the QFT'd MPO. The MPO-mode `get_bands` pipeline must *not* be used here — the exciton Chebyshev-MPO bond dimension explodes. All momenta are 1-indexed $(1:2^L)$, LSB-first, with electron bits on odd sites and hole bits on even sites.

Momentum-basis probe states. Compact finite-state MPS on the interleaved $2L$ -qubit chain, each built in $O(L)$ via a carry bond that enforces $k + h \equiv Q \pmod{N}$, never by explicit summation over the $N = 2^L$ momenta. `mpsexcitonQ`, `mpsexcitonQTrace`, and `mpsexcitonKQ` now live in **Two-Particle Systems (twoparticle.tk.jl) → MPS Basis-State Probes**, alongside the real-space `mpsexciton` probes.

Spectral drivers. MPS-KPM analogues of `get_bands`: one online Chebyshev recursion per momentum (or averaged group), accumulating all energies in a single pass. `H_QFT` supplies the conjugated MPO; the real-space `H::TBHamiltonian` supplies metadata (`sites`, `N`, `L`, `scale`, `center`). Rows are energies, columns are total momenta.

Signature	Description
<code>get_exciton_bands(H_QFT, H, Ncheb, ω_phys_vals; Q_list, Q_groups, ...) → Matrix{Float64}</code>	Coherent exciton bands $A(Q, \omega)$ from the <code>mpsexcitonQ</code> pair state. <code>Q_list</code> selects momenta; <code>Q_groups</code> averages several probes into one column; otherwise a window $(q_start/q_end/num_q/num_avg)$ is sampled. The <code>K_*/k_*</code> kwargs are backward-compatible aliases.
<code>get_exciton_continuum(H_QFT, H, Ncheb, ω_phys_vals; Q_list, N_sample, k_list, ...) → Matrix{Float64}</code>	Incoherent continuum $A_{\text{cont}}(Q, \omega) = N^{-1} \sum_k \langle k, Q-k \delta(\omega - H) k, Q-k \rangle$ via <code>mpsexcitonQTrace</code> random-phase traces (or a deterministic <code>k_list</code>). <code>N_sample</code> sets the number of trace probes per Q . <code>normalize=true</code> (default) returns the per-relative-momentum average; <code>false</code> returns the total fixed- Q trace ($\times H.N$).

Online Band Structure: `get_bands`

get_bands runs an online Chebyshev KPM recurrence over the full MPO space without ever materialising the full T_n list. It replaces the legacy inner-product approach and requires no prior `KPM.Tn` call.

Signature	Description
<code>get_bands(H_mpo, scale, center, sites, Ncheb, D, ω_vals; ...) → Matrix{Float64}</code>	Low-level form. ω _vals must be in rescaled units $\in (-1, 1)$. <code>D=1</code> or <code>2</code> sets the spatial dimension. Returns $N_\omega \times N_k$ matrix.
<code>get_bands(H::TBHamiltonian, Ncheb, D, ω_phys_vals; ...) → Matrix{Float64} or NamedTuple</code>	High-level form. Physical energies rescaled via <code>H.scale/H.center</code> . All auxiliary indices auto-detected from struct fields; no manual index passing. Returns <code>NamedTuple(Ak, ticks, labels)</code> when <code>kpath</code> kwarg is used, otherwise <code>Matrix{Float64}</code> .
<code>get_bands(H::TBHamiltonian, Ncheb, ω_phys_vals; ...) → Matrix{Float64} or NamedTuple</code>	Convenience form; infers <code>D</code> from <code>length(H.geometry(1))</code> . Errors if <code>H.geometry</code> is <code>nothing</code> .

Five-step projection pipeline (all steps optional and composable):

#	Step	Kwargs	Notes
0	Nambu (BdG)	<code>nambu_proj</code> ,	Sectors 1=particle, 2=hole.
1	Spin	<code>proj_nambu</code>	
		<code>spin_proj</code> ,	Sectors 1= $\uparrow$, 2= $\downarrow$.
		<code>proj_s</code> , <code>spin_s_aux</code>	
1c	Layer	<code>layer_proj</code> ,	For bilayer/multilayer models.
		<code>proj_layer</code>	
1b	Sublattice (aux)	<code>sublat_proj</code> ,	For kagomé, Lieb, honeycomb, dice
		<code>proj_sl</code> , <code>sublat_s</code>	(<code>H.sublattice_s</code> set).
2	Sublattice (legacy)	<code>sublattice</code> ,	Mask sandwich for preset models
		<code>proj_sl</code>	without <code>sublattice_s</code> .

`proj_*=nothing` sums over all sectors; `proj_*=k` selects sector k .

k-point sampling kwargs: `xmin`, `xmax`, `num_x` (grid in k_x); `ymin`, `ymax`, `num_y` (2D); `num_avg` (points averaged per center); `k_groups_override` (explicit path from `kpath_2d`).

High-symmetry k-path shortcut:

```
res = get_bands(H, Ncheb, 2, omega;
               kpath=[:G, :M, :Kp, :G], kpath_lattice=:honeycomb, num_x=30)
# res.Ak -- (N_omega x N_k) matrix
# heatmap(1:size(res.Ak,2), omega, res.Ak; xticks=(res.ticks, res.labels))
```

High-Symmetry k-Path Utilities

Signature	Description
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<code>hsk_honeycomb(Lx, Ly) → NamedTuple</code>	High-symmetry k-points for the honeycomb sublattice Hamiltonian in quantics integer units. Fields: <code>G=(0,0)</code> , <code>M=(Nx/2,Ny/4)</code> , <code>K</code> , <code>Kp</code> (Dirac points, rounded to nearest integer).
<code>hsk_square(Lx, Ly) → NamedTuple</code>	High-symmetry points for square lattice: <code>G</code> , <code>X=(Nx/2,0)</code> , <code>M=(Nx/2,Ny/2)</code> .
<code>hsk_triangular(Lx, Ly) → NamedTuple</code>	Approximate high-symmetry points for triangular lattice: <code>G</code> , <code>M</code> , <code>K</code> .
<code>kpath_2d(hs_points, Lx; npts_per_segment=20) → (k_groups, tick_positions)</code>	Interpolate between <code>(kx_idx,ky_idx)</code> tuples; returns <code>k_groups</code> for <code>k_groups_override</code> and <code>tick_positions</code> for plot axes.
<code>kpath_setup(lattice, Lx, Ly, path_syms; npts_per_segment=20) → (k_groups, ticks, labels)</code>	Build k-path from symbol list (e.g. <code>[:G,:M,:Kp,:G]</code>) for given lattice (<code>:honeycomb</code> , <code>:square</code> , <code>:triangular</code>). Labels use Unicode: <code>G</code> → Γ , <code>Kp</code> → K' . Called internally by <code>get_bands</code> <code>kpath</code> shortcut.

Auxiliary Index Projection Helpers

Signature	Description
<code>project_aux(W, aux_s, sigma; side=:pre) → MPO</code>	Remove one auxiliary site from L -site MPO <code>W</code> by projecting onto state σ . Returns $(L-1)$ -site position-only MPO. <code>side=:pre</code> (prepended) or <code>:post</code> (postpended).
<code>project_spin(W, spin_s, sigma) → MPO</code>	Convenience alias for <code>project_aux(W, spin_s, sigma; side=:pre)</code> .
<code>aux_site(H::TBHamiltonian, which) → (Index, Symbol)</code>	Return the auxiliary Index and its side (<code>:pre</code> / <code>:post</code>) for <code>which</code> $\in$ <code>{:spin,:nambu,:layer,:sublattice}</code> . Auto-called by the <code>TBHamiltonian</code> overload of <code>get_bands</code> .

Note: `get_spect_k` and `get_spect_k_doubled` (inner-product approach) have been superseded by the diagonal-extraction approach in `get_bands` and are no longer part of the public API.

Topological Invariants (Topology_tk.jl)

Real-space topological invariants (Chern number and winding number) for general lattice geometries. All functions accept user-supplied coordinate functions `xfunc(i, L_chain)` and `yfunc(i, L_chain)` where `i` is a 0-indexed site number and `L_chain=2l` is the relevant lattice length.

Key concept — quenching: Periodic boundary conditions would make $\hat{x}$ discontinuous. With `quenched=true` the position operators are replaced by $\sin(x/\Lambda)$ and $\cos(x/\Lambda)$, which are bounded and smooth. The identity $x \approx \Lambda \sin(x/\Lambda)$ for small arguments means the quenched Chern marker recovers physical units via a factor Λ^2 . The winding-number density recovers via Λ .

Projector Retrieval

Signature	Description
<code>_get_projector(H; method=:KPM, fermi=0.0, Nchebychev=300, maxdim=40, cutoff=1e-8, Nel=nothing) → MPO</code>	Compute or retrieve the zero-temperature projector $P = \theta(\mu - H)$. <code>method=:KPM</code> : uses cached <code>H._tn_cache</code> or runs KPM; <code>fermi</code> in physical units. <code>method=:mcweeny</code> : uses <code>H._density_cache</code> or calls <code>mcweeny_purify</code> . <code>method=:sp2</code> : uses <code>H._density_cache</code> or calls <code>sp2_purify</code> with target <code>Nel</code> .

Quenched Position Operators

These build diagonal MPOs for the trigonometric quenching functions. The argument to `get_diagonal_mpo` is 1-indexed; these builders convert internally with `xfunc(i-1, L_chain)`.

Signature	Description
<code>get_sinx_op(L, sites, L_chain, Λ, xfunc) → MPO</code>	Diagonal MPO for $\sin(x_i/\Lambda)$.
<code>get_cosx_op(L, sites, L_chain, Λ, xfunc) → MPO</code>	Diagonal MPO for $\cos(x_i/\Lambda)$.
<code>get_siny_op(L, sites, L_chain, Λ, yfunc) → MPO</code>	Diagonal MPO for $\sin(y_i/\Lambda)$.
<code>get_cosal_op(L, sites, L_chain, Λ, yfunc) → MPO</code>	Diagonal MPO for $\cos(y_i/\Lambda)$.

1D Winding Number

Signature	Description
<code>get_W(H::TBHamiltonian, xfunc=nothing; method=:KPM, fermi=0.0, Nchebychev=300, maxdim=15, cutoff=1e-8, Nel=nothing, quenched=true, l=nothing, $\Lambda=10$) → Function</code>	Compute the winding-number density operator. <code>H</code> must have a 2-component sublattice index (<code>H.sublattice_s</code> , dim 2); $\sigma_z = \text{diag}(+1, -1)$ on that index is built automatically. <code>xfunc(i, L_chain)</code> (0-indexed) defaults to <code>nothing</code> , auto-derived from <code>H.geometry_uc</code> (preferred) or <code>H.geometry</code> . Pre-computes one or two MPOs (W_1, W_2 for <code>quenched=true</code>). Returns a closure <code>calculate_winding(uc::Int) -> ComplexF64</code> where <code>uc</code> is a 1-indexed unit-cell number ($1 \dots 2^L$); the value sums the winding marker over both sublattice atoms (A and B) in that cell.

Quenched case (`quenched=true`): Uses the 2-term decomposition $\langle \alpha | \hat{W} | \alpha \rangle = \Lambda [\cos(x_\alpha/\Lambda) \langle \alpha | W_1 | \alpha \rangle - \sin(x_\alpha/\Lambda) \langle \alpha | W_2 | \alpha \rangle]$ where $W_1 = \sigma_z(P \sin \hat{X} Q + Q \sin \hat{X} P)$, $W_2 = \sigma_z(P \cos \hat{X} Q + Q \cos \hat{X} P)$, and α ranges over the two sublattice sites (A, B) of unit cell `uc`.

Flat case (`quenched=false`): builds $W = \sigma_z(P\hat{x}Q + Q\hat{x}P)$ once; the closure evaluates $\sum_{\alpha \in \{A,B\}} \langle \alpha | W | \alpha \rangle$ for unit cell `uc`.

2D Real-Space Chern Marker

Signature	Description
<code>get_C_op_MPO_from_P(P, L, sites, xfunc, yfunc; l=nothing, $\Lambda=10$, maxdim=500, cutoff=1e-8, quenched=true, sequential=false, pk_mpo=nothing) → Function</code>	Build real-space Chern marker from pre-computed projector P . <code>sequential=true</code> avoids forming the 4 α -independent C_1 – C_4 MPOs, instead applying operators directly to each basis MPS (fewer MPO-MPO products, more inner products per site). <code>pk_mpo</code> optionally sandwiches a projector (e.g. a valley projector from <code>get_valley_C</code>) into the marker: $\langle \alpha P_K C P_K \alpha \rangle$. Returns closure <code>calculate_chern_number(uc::Int) → ComplexF64</code> , <code>uc</code> a 1-indexed unit-cell number ($1 \dots 2^L$), summed over sublattice atoms.
<code>get_C(H::TBHamiltonian, xfunc=nothing, yfunc=nothing; method=:KPM, fermi=0.0, l=nothing, $\Lambda=10$, Lambda=nothing, Nchebychev=300, maxdim=500, cutoff=1e-8, Nel=nothing, quenched=true, sequential=false) → Function</code>	High-level wrapper: calls <code>_get_projector</code> then <code>get_C_op_MPO_from_P</code> . <code>xfunc/yfunc</code> default to <code>nothing</code> and are auto-derived from <code>H.geometry_uc</code> (sublattice models) or <code>H.geometry</code> . <code>Lambda</code> is an ASCII alias for Λ . Returns same <code>calculate_chern_number</code> closure.

Quenched case (`quenched=true`): uses the 4-term decomposition

$$\begin{aligned} \langle \alpha | \hat{C} | \alpha \rangle = 2\pi i \Lambda^2 & \left[\cos \frac{x_\alpha}{\Lambda} \cos \frac{y_\alpha}{\Lambda} \langle \alpha | C_1 | \alpha \rangle + \sin \frac{x_\alpha}{\Lambda} \sin \frac{y_\alpha}{\Lambda} \langle \alpha | C_2 | \alpha \rangle \right. \\ & \left. - \cos \frac{x_\alpha}{\Lambda} \sin \frac{y_\alpha}{\Lambda} \langle \alpha | C_3 | \alpha \rangle - \sin \frac{x_\alpha}{\Lambda} \cos \frac{y_\alpha}{\Lambda} \langle \alpha | C_4 | \alpha \rangle \right] \end{aligned}$$

where C_1 – C_4 are pre-computed α -independent MPOs. Only scalar trig evaluations are needed per site.

Flat case (`quenched=false`): $\hat{C} = 2\pi i(Q\hat{x}P\hat{y}Q - P\hat{x}Q\hat{y}P)$.

Valley Operator and Valley Chern Number (Honeycomb)

Signature	Description
<code>get_valley_operator(H::TBHamiltonian; maxdim=500, cutoff=1e-8) → MPO</code>	Build the valley operator V for a 2D honeycomb Hamiltonian (<code>H.Lx</code> set, 2-component <code>H.sublattice_s</code>). V is the Haldane next-nearest-neighbour Hamiltonian ($\varphi = \pi/2$, NN term zeroed) multiplied by a sublattice sign η_i (+1 on A, −1 on B), with global prefactor $-i/(3\sqrt{3})$.

<pre>get_valley_projectors(V_mpo::MPO, sites; maxdim=500, cutoff=1e-8) → (PK, PK_prime) get_valley_C(H::TBHamiltonian, xfunc=nothing, yfunc=nothing; valley=:K, use_sign=true, method=:mcweeny, fermi=0.0, l=nothing, Λ=10, Nchebychev=300, maxdim=500, cutoff=1e-8, Nel=nothing, quenched=true, sequential=false) → Function</pre>	Return the two valley projectors $P_K = (I + V)/2$ and $P_{K'} = (I - V)/2$ from valley operator V_{mpo}. Valley-resolved Chern marker. Builds $V = \text{get_valley_operator}(H)$; if <code>use_sign=true</code> (default) sharpens it to $S = \text{sign}(V) \in \{-1, +1\}$ via <code>sign_mpo</code>. Forms $P_K = P(I \pm S)/2$ (<code>sign +</code> for <code>valley=:K</code>, <code>-</code> for <code>:K_prime</code>), symmetrizes it, and passes it as <code>pk_mpo</code> to <code>get_C_op_MPO_from_P</code>, which evaluates $\langle \alpha P_K C P_K \alpha \rangle$. Returns closure <code>calculate_valley_chern(uc::Int) -> ComplexF64</code>.
---	--

Thouless Pump

Computes the Thouless pump Chern number via the M1Q marker: $C = \text{M1Q}(T) - \text{M1Q}(0)$, where $\text{M1Q}(r, t) = \langle r | P(t) U^\dagger(t) \hat{x} U(t) P(t) | r \rangle$ and $U(t)$ is the adiabatic evolution operator propagated by $h(t) = [\dot{P}, P]$ via a second-order Taylor step.

Signature	Description
<pre>get_pump_xop(L, sites, xfunc; quenched=false, Λ=-1.0) → MPO</pre>	Build the position operator MPO for Thouless pump. <code>quenched=false</code>: diagonal entries are <code>xfunc(i, N)</code> directly. <code>quenched=true</code>: entries are $\Lambda \sin(x/\Lambda)$ to smooth the PBC discontinuity (Λ defaults to $N = 2^L$).
<pre>thouless_pump(P_array, dt, x.op, sites; r_center, maxdim=100, cutoff=1e-8, verbose=false, return_trajectory=false) → Float64 or (Float64, Vector{Float64})</pre>	Compute C from a pre-built array of projectors <code>P_array</code> (length <code>Nt</code>). <code>r_center</code>: 0-indexed bulk site at which M1Q is evaluated. With <code>return_trajectory=true</code> returns $(C, \text{M1Q_traj})$ where <code>M1Q_traj</code> has length <code>Nt+1</code> (index $1 = t = 0$).
<pre>get_thouless_pump(H_of_t, Nt, T, xfunc; P_method=:mcweeny, fermi=0.0, Nchebychev=200, maxdim=100, cutoff=1e-8, quenched=false, Λ=-1.0, Nel=nothing, r_center=nothing, verbose=false) → Float64</pre>	High-level wrapper. Calls <code>H_of_t(t_k)</code> to obtain the Hamiltonian at each of the <code>Nt</code> time steps, builds $P(t_k)$ via <code>P_method</code> (<code>:mcweeny</code>, <code>:sp2</code>, or <code>:KPM</code>), then calls <code>thouless_pump</code>. <code>r_center</code> defaults to $2^L/2$.

Typical usage:

```
# Factory: same site indices must be reused for all t
H0 = get_Hamiltonian("chain_1d", 1.0; L=8)
ref_sites = H0.sites
H_of_t(t) = get_Hamiltonian("ssh", (t=1.0, d=0.5*cos(2*pi*t/T));
                             L=8, ref_sites=ref_sites)

xfunc(i, N) = Float64(div(i, 2)) # x-coord for SSH (two sites per unit cell)
C = get_thouless_pump(H_of_t, 40, T, xfunc;
```

```

P_method=:mcweeny, verbose=true)
println("Thouless pump invariant: ", round(C; digits=3))

```

DMRG (DMRG_tk.jl)

Signature	Description
<code>dmrg_gs(H, sites; nsweeps=10, maxdim=[10,20,50,100,200], cutoff=1e-8, noise=[...], linkdim_init=10) → (E, ψ)</code>	Ground state and energy of H via DMRG.
<code>build_K(H_mpo, sites, ω, η; maxdim_K=200, cutoff_K=1e-8) → MPO</code>	Resolvent-squared MPO $K(\omega, \eta) = (H - \omega I)^2 + \eta^2 I$. Ground-state energy $E_K \geq \eta^2$; $E_K \rightarrow \eta^2$ at spectral peaks.
<code>dmrg_spectral(H_mpo, sites, ω, η; ψ_0=nothing, ...) → (E_K, ψ)</code>	DMRG ground state of $K(\omega, \eta)$. Output $ \psi(\omega)\rangle$ concentrates on the eigenstate of H nearest ω .
<code>local_weight(ψ, i, L, sites) → Float64</code>	$ \langle i \psi\rangle ^2$: LDoS proxy at site i (0-indexed, big-endian). Use with <code>dmrg_spectral</code> output.

Time Evolution (Timeev_tk.jl)

Propagator MPO and Single-Step TDVP

Signature	Description
<code>tdvp_evolve(H, ψ, dt; maxdim=200, cutoff=1e-10, normalize=true, nsite=2) → MPS</code>	Single TDVP step; thin wrapper around <code>ITensorMPS.tdvp</code> .
<code>build_tdvp_propagator_mpo(H, dt, L, sites; maxdim=50, cross_tol=1e-8, ...) → MPO</code>	Compress $U(\Delta t) = e^{-iH\Delta t}$ into an MPO via QTCI on matrix elements $\langle i U j\rangle$ evaluated by TDVP. Diagonal pivots seeded by default. <code>TBHamiltonian</code> overload available.
<code>apply_mpo_to_mps(U_mpo, ψ; cutoff=1e-12, maxdim=500, normalize=true) → MPS</code>	Apply propagator MPO to a single MPS state; optionally normalize.

Trajectory Evolution: MPS States

Signature	Description
<code>evolve_with_propagator(U_mpo, ψ_0, nsteps; normalize_each_step=true, cutoff=1e-8, maxdim=10000) → Vector{MPS}</code>	Apply the same propagator MPO <code>nsteps</code> times; returns length-(<code>nsteps</code> +1) trajectory including the initial state.
<code>evolve_with_tdvp(H, ψ_0, nsteps, dt; normalize_each_step=true, maxdim=200, cutoff=1e-10, ...) → Vector{MPS}</code>	Multi-step TDVP trajectory for time-independent Hamiltonian. Returns full trajectory [<code>psi(0)</code> , <code>psi(dt)</code> , ..., <code>psi(nsteps*dt)</code>]. <code>TBHamiltonian</code> overload: uses $-iH.mpo$.

```
evolve_with_tdvp_timedep(Hoft,  $\psi_0$ ,
nsteps, dt; normalize_each_step=true,
maxdim=200, cutoff=1e-10,
reverse_step=false, krylovdim=20,
tol=1e-10) → Vector{MPS}
```

TDVP trajectory for a time-dependent $H(t)$. Uses midpoint rule: H is frozen at $H(t + \Delta t/2)$ over each interval. `Hoft` is a callable `t → MPO` returning $H(t)$ itself (the function applies $-iH$ internally); no `TBHamiltonian` overload is provided for this function.

State Analysis Utilities

Signature	Description
<code>compute_basis_overlaps(states, L, sites) → NamedTuple</code>	Overlap of each MPS in <code>states</code> with all 2^L computational basis states. Returns (<code>overlaps</code> , <code>abs_overlaps</code> , <code>probabilities</code> , <code>norms</code>).
<code>basis_amplitude(ψ, n, L, sites) → ComplexF64</code>	Single-basis-state amplitude $\langle n \psi\rangle$ (0-indexed <code>n</code>).
<code>phase_aligned_distance(ψ_a, ψ_b) → Float64</code>	$\sqrt{2 - 2 \langle \hat{\psi}_a \hat{\psi}_b\rangle }$: distance minimised over global phase; 0 if states are equal up to phase and normalisation.
<code>check_tdvp_vs_U_mpo(H, U_mpo, dt, L, sites; ...) → (Float64, Float64)</code>	Validate <code>U_mpo</code> against direct TDVP on computational basis states. Returns (<code>max_overlap_error</code> , <code>max_distance</code>). <code>TBHamiltonian</code> overload available.

Density Matrix Dynamics (open / driven systems)

The density matrix ρ is represented as an MPO. The von Neumann equation $d\rho/dt = -i[H(t), \rho]$ is integrated with a classical RK4 scheme. Each $H(t)$ evaluation produces one MPO-MPO product, so bond dimension grows; use `maxdim` and `cutoff` to control cost.

Signature	Description
<code>rk4_step_dm_timedep(Hoft, ρ, t, dt; maxdim=200, cutoff=1e-10, truncate_intermediates=true) → MPO</code>	Single RK4 step for $d\rho/dt = -i[H(t), \rho]$. Evaluates H at the four RK4 sample times.
<code>evolve_rk4_dm_timedep(Hoft, ρ_0, nsteps, dt; maxdim=200, cutoff=1e-10, verbose=false) → Vector{MPO}</code>	Full RK4 trajectory; returns length-(<code>nsteps+1</code>) list including ρ_0 .
<code>dm_expect(O, ρ) → Float64</code>	Expectation value $\text{Tr}(O\rho)$ for Hermitian <code>O</code> . Uses <code>inner(O, rho)</code> .
<code>purity(ρ) → Float64</code>	$\text{Tr}(\rho^2) = \text{inner}(\rho, \rho)$. 1 for pure states, < 1 for mixed.
<code>purity_trajectory(states) → Vector{Float64}</code>	Apply purity to each MPO in <code>states</code> .

Observables Along Trajectories

Signature	Description
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<code>observables_trajectory(ops, states::Vector{MPO}) → Dict</code>	Evaluate a named collection of MPO observables on a density-matrix trajectory. <code>ops</code> is a <code>DictString,MPO</code> ; returns <code>DictString,VectorFloat64</code> .
<code>timedep_observable_trajectory(OfT, states::Vector{MPO}, dt) → Vector{Float64}</code>	Evaluate a time-dependent observable <code>OfT(t) → MPO</code> at times $t_k = k \cdot \Delta t$ on the density-matrix trajectory.
<code>bond_current_x(ρ, j, tx, L, sites) → ComplexF64</code>	Bond current $J_j = 2 \text{Im}[t_x \rho_{j,j+1}]$ on the bond $(j, j+1)$ (0-indexed <code>j</code>). Uses off-diagonal matrix elements of ρ via <code>matrix_checker</code> .
<code>bond_current_x_trajectory(states, j, tx, L, sites; dt=1.0) → Vector{ComplexF64}</code>	Evaluate <code>bond_current_x</code> at every step of a trajectory.
<code>central_x_bond(L; Nx=nothing) → Int</code>	Return the 0-indexed site index of the central x -bond. For 2D ($L_x = L_y = L/2$) provides the horizontal bond nearest the geometric center; uses L_x rows if <code>Nx</code> is given.

Diagnostic Tools

Signature	Description
<code>compare_propagator_and_tdvp_heatmaps(U_mpo, H, ψ_0, L, sites, nsteps; ...) → NamedTuple</code>	Run both propagator-MPO and TDVP trajectories from the same initial state and return basis-state probability heatmaps for side-by-side comparison. <code>TBHamiltonian</code> overload available.

Two-Particle Systems (twoparticle_tk.jl)

Constructs the exciton (electron-hole) Hamiltonian on an interleaved $2L$ -site space.

Site encoding: L position qubits per sector, interleaved as $[e_1, h_1, e_2, h_2, \dots]$: odd sites $(1, 3, \dots)$ = electron, even sites $(2, 4, \dots)$ = hole. The Hamiltonian is $H_{\text{exc}} = H_c \otimes I_h - I_e \otimes H_v + U$ where U is a contact interaction diagonal in the electron-hole coincidence basis.

`TBHamiltonian.L` = position qubits per sector; `TBHamiltonian.sites` holds all $2L$ interleaved sites; `TBHamiltonian.N` = 2^L (single-sector physical sites).

High-Level Constructors

Signature	Description
<code>exciton_hamiltonian(geometry, params, Ufunc; L, on_site=nothing, scale=nothing, tol_quantics=1e-8, maxbonddim_quantics=100, tol=1e-8, cutoff=1e-8, maxdim=200, kwargs...) → TBHamiltonian</code>	Build an exciton Hamiltonian where both electron and hole sectors share the same <code>geometry/params</code> . Extra <code>kwargs</code> are forwarded to <code>get_Hamiltonian</code> .

<pre>exciton_hamiltonian(H_c, H_v, Ufunc; on_site=nothing, scale=nothing, tol_quantics=1e-8, maxbondddim_quantics=100, tol=1e-8, cutoff=1e-8, maxdim=200) → TBHamiltonian</pre>	Two-sector form for distinct electron/hole bands. <code>H_c</code> and <code>H_v</code> must have the same <code>L</code> and distinct site indices.
---	---

Keyword arguments (both forms):

- `Ufunc(x)` — contact interaction strength at site $x \in \{1, \dots, 2^L\}$ (1-indexed).
- `on_site` — conduction-band-edge modulation $V(x)$; added as $+V$ on electron, $-V$ on hole (type-I confinement: both carriers confined). Compressed via QTCL.
- `scale` — exciton half-bandwidth; 0.0 triggers lazy DMRG estimation on first KPM call.

Low-Level MPO Builders

Signature	Description
<code>Exciton_Hamiltonian(H_c, H_v, Ufunc; on_site, tol_quantics, maxbondddim_quantics, tol, cutoff, maxdim) → MPO</code>	Low-level builder returning a raw $2L$ -site MPO. Interleaves electron/hole kinetic terms and adds contact interaction.
<code>hopping_mpo_exciton(H_c, H_v) → MPO</code>	Embed single-particle sectors into the $2L$ -site space: $H_c \otimes I_h - I_e \otimes H_v$ without interaction.
<code>build_interaction_op_exciton(L, sites, Ufunc) → MPO</code>	Build the contact interaction diagonal MPO. <code>Ufunc(x)</code> on the diagonal where electron and hole coincide ($x_e = x_h = x$); off-diagonal elements are zero.

Typical usage:

```
# 1D exciton -- same geometry, contact interaction
H_exc = exciton_hamiltonian("chain_1d", 1.0, x -> -0.5;
                           L=8, on_site = x -> -0.1*exp(-((x-64)^2)/200))

# 2D exciton -- separate electron/hole bands
H_c = get_Hamiltonian("square_2d", 1.0; L=8, Lx=4, Ly=4)
H_v = get_Hamiltonian("square_2d", 0.5; L=8, Lx=4, Ly=4)
H_exc = exciton_hamiltonian(H_c, H_v, x -> -0.2)

# Use with standard TensorBinding tools
Tn, scale, center = KPM_Tn(H_exc, 200)
dos = real(tr(get_ldos_w_from_Tn(Tn, 200, 0.0; maxdim=200)))
```

MPS Basis-State Probes

Real-space and momentum-space MPS probes on the interleaved $2L$ -site electron-hole chain. The real-space probes are plain product states, 1-indexed ($1:2^L$), MSB-first (matching `get_diagonal_mpo`). The momentum-space probes are compact finite-state MPS built in $O(L)$ via a carry bond enforcing $k+h \equiv Q \pmod{N}$, never by explicit summation over the $N = 2^L$ momenta; momentum indices are 1-indexed ($1:2^L$), LSB-first, with electron bits on odd sites and hole bits on even sites.

Signature	Description
<code>mpsexciton(xe, xh, sites) → MPS</code>	Real-space product state $ x_e, x_h\rangle$: <code>xe/xh</code> are 1-indexed electron/hole positions in $1:2^L$, odd sites = electron qubit, even sites = hole qubit. Used by <code>get_exciton_ldos_separation</code> .
<code>mpsexciton(x, sites) → MPS</code>	<code>= mpsexciton(x, x, sites)</code> : the bound $ x, x\rangle$ probe ($d = 0$ electron-hole separation), used by <code>get_exciton_ldos_spatial</code> .
<code>mpsexcitonQ(Q, sites) → MPS</code>	Coherent fixed-total-momentum pair state $ Q\rangle = N^{-1/2} \sum_k k, -k + Q\rangle$. The reference probe for exciton <i>bands</i> (<code>get_exciton_bands</code>).
<code>mpsexcitonQTrace(Q, sites; rng) → MPS</code>	Random-phase (± 1) trace probe $ r_Q\rangle = N^{-1/2} \sum_k \eta_k k, Q - k\rangle$ with $E[\eta_k \eta_l] = \delta_{kl}$, so averaging probes estimates the fixed- Q continuum trace. The stochastic <i>continuum</i> probe (<code>get_exciton_continuum</code>).
<code>mpsexcitonKQ(k, Q, sites) → MPS</code>	Single product basis vector $ k, Q - k\rangle$ in the fixed- Q sector; the deterministic continuum probe sampled by <code>get_exciton_continuum</code> .

Random Phase Approximation (RPA_tk.jl)

Computes the RPA susceptibility χ^{RPA} via the Dyson equation $(1 - \Pi_0 V)\chi = \Pi_0$, where Π_0 is the non-interacting (Lindhard) polarization bubble. The high-level `TBHamiltonian` API (`get_bubble_mpo`, `get_rpa_susceptibility`, and the magnon variants) is the recommended entry point; the legacy `get_Tnlists` / `build_bubble_mpo` pipeline remains available as a lower-level building block.

High-Level TBHamiltonian API

Signature	Description
<code>get_bubble_mpo(H1::TBHamiltonian, H2::TBHamiltonian, ω; $\epsilon_F=0.0$, <code>P_method=:purification</code>, <code>GF_method=:kpm</code>, <code>Ncheb=150</code>, <code>maxdim=200</code>, <code>cutoff=1e-8</code>, <code>purify_method=:mcweeny</code>, <code>purify_maxdim=40</code>, <code>purify_maxiters=30</code>, <code>purify_tol=1e-5</code>, $\eta=1e-3$, <code>krylov_nsweeps=12</code>, <code>krylov_maxdim=100</code>, <code>krylov_cutoff=1e-8</code>, <code>verbose=false</code>) → MPO</code>	Non-interacting polarization bubble $\Pi_0(\omega)$ on <code>H1.sites</code> . <code>P_method=:purification</code> reuses <code>H.density_cache</code> when present; <code>P_method=:kpm</code> computes P via <code>get_density_from_Tn</code> . <code>GF_method=:kpm</code> (auto spectral-bound estimation) or <code>:krylov</code> (via <code>get_green_krylov</code>) for $G_{\text{eff}}(\omega)$, $H_{\text{eff}} = \mathbf{I} \otimes H_2 - H_1 \otimes \mathbf{I}$.

`get_rpa_susceptibility(H::TBHamiltonian, RPA susceptibility $\chi(\omega)$, diagonal χ_{ijjj}^{RPA} on MPOV, ω ; mode=:charge, a $2L$ -site MPS. mode=:charge (default) rpa_nsweeps=20, rpa_maxdim=400, computes the density-density bubble via rpa_cutoff=1e-8, (get_bubble_mpo get_bubble_mpo(H,H, ω); mode=:magnetic projects H onto its spin- $\uparrow/\downarrow$ blocks and calls get_magnon_bubble (requires H.spin_s != nothing). Solves $(I - \Pi_0 V)\chi = \Pi_0$ via rpa_from_bubble_diag.` `kwargs`) $\rightarrow$ MPS

Magnon (Transverse Spin) Susceptibility

Signature	Description
<code>get_magnon_bubble(H::TBHamiltonian, ω; (get_bubble_mpo kwargs)) $\rightarrow$ MPO</code>	Transverse spin bubble $\Pi_0^{+-}(\omega) = \sum_k (f_{k\downarrow} - f_{k\uparrow}) / (\omega - (\varepsilon_{k\uparrow} - \varepsilon_{k\downarrow}) + i\eta)$. Projects H onto spin- $\uparrow/\downarrow$ blocks via <code>_project_spin_sector</code> and calls <code>get_bubble_mpo(H.up,H.dn,ω)</code> . Errors if H.spin_s == nothing.
<code>get_magnon_susceptibility(H::TBHamiltonian, MPOV, ω; rpa_nsweeps=20, rpa_maxdim=400, rpa_cutoff=1e-8, (get_bubble_mpo kwargs)) $\rightarrow$ MPS</code>	RPA transverse spin susceptibility $\chi_{\text{RPA}}^{+-}(\omega)$. Builds $\Pi_0^{+-}(\omega)$ via <code>get_magnon_bubble</code> , then solves $(I - \Pi_0 V)\chi = \Pi_0$. For a Hubbard-like interaction, MPOV = $U \cdot \text{Id}$.
<code>get_magnon_susceptibility_wynn(H::TBHamiltonian, MPOV, ωlist; K_max=6, maxdim_apply=200, cutoff_apply=1e-8, verbose=false, (get_bubble_mpo kwargs)) $\rightarrow$ (chi.partial, chi.wynn)</code>	Wynn-accelerated transverse spin RPA susceptibility over a frequency list. Spin- $\uparrow/\downarrow$ projections are done once outside the ω loop; otherwise identical to <code>get_rpa_susceptibility_wynn</code> .

Wynn ε -Accelerated RPA (Neumann Series)

Instead of solving $(I - \Pi_0 V)\chi = \Pi_0$ directly, builds the Neumann series $T_0 = \Pi_0$, $T_n = T_{n-1} \cdot V \cdot \Pi_0$ (so $\sum_n T_n \rightarrow \chi^{\text{RPA}}$ as $K \rightarrow \infty$), extracts k -space scalars $s_n(q, \omega) = -\text{Im}\langle q | T_n(\omega) | q \rangle$ via `get_spect_k`, and applies the Wynn ε -algorithm to the partial-sum sequence for fast convergence.

Signature	Description
<code>wynn_epsilon(s) $\rightarrow$ Vector{ComplexF64}</code>	Wynn ε -algorithm on a scalar sequence $s = [s_0, \dots, s_K]$. Returns even-column first-row Padé estimates $[\varepsilon_2(0), \varepsilon_4(0), \dots]$. Uses only additions and reciprocals.
<code>get_spect_k(W::MPO; tol=1e-9, maxdim=100) $\rightarrow$ Vector{ComplexF64}</code>	k -space diagonal of MPO W as a dense length- 2^L vector. Conjugates W by the QFT, extracts the diagonal as an MPS, and evaluates $\langle k \tilde{W} k \rangle$ for all 2^L quantics k -indices (LSB-first convention).

`get_rpa_susceptibility_wynn(H::TBHamiltonian,  $\omega$ list; mode=:charge, K_max=6, maxdim_apply=200, cutoff_apply=1e-8, verbose=false, (get_bubble_mpo kwargs))` $\rightarrow$ (`chi_partial`, `chi_wynn`)
`rpa_wynn_from_bubbles( $\Pi_0$ _list, MPOV; K_max=6, maxdim_apply=200, cutoff_apply=1e-8, verbose=false)` $\rightarrow$ (`chi_partial`, `chi_wynn`)

`get_rpa_susceptibility_wynn` computes $\chi(q, \omega)$ for all ω in ω list via Wynn Padé acceleration of the bubble series. `mode=:charge` (default) uses `get_bubble_mpo(H, H,  $\omega$ )`; `mode=:magnetic` uses `get_magnon_bubble` (requires `H.spin_s != nothing`). `chi_partial[k+1, i $\omega$ , q]` = partial sum $\sum_{n=0}^k s_n(q, \omega)$; `chi_wynn[m, i $\omega$ , q]` = $\varepsilon_{2m}(0)$ estimate. Same as `get_rpa_susceptibility_wynn` but takes a precomputed `Vector{MPO}` of bubbles directly (e.g. from `get_bubble_mpo_cheb2d_tucker` or `get_bubble_mpo_haydock`), skipping the internal bubble calculation.

Haydock Recursion (Operator-Level Krylov)

An alternative to KPM/linsolve for $G_{\text{eff}}(\omega)$: builds an orthogonal Krylov basis $\{\Phi_n\}$ of MPOs under H_{eff} via Lanczos recursion with the Frobenius inner product $(A, B) = \text{Tr}[A^\dagger B]$, then evaluates the resolvent for any ω by solving the small tridiagonal Lanczos system — the expensive recursion runs once and is reused for every frequency.

Signature	Description
<code>haydock_cf(H_mpo, seed, N_steps; maxdim=200, cutoff=1e-8, verbose=false)</code> $\rightarrow$ (<code>a</code> , <code>b</code> , <code>basis</code> , <code>norm0</code>)	Lanczos recursion $\Phi_{n+1} = H \cdot \Phi_n - a_n \Phi_n - b_n \Phi_{n-1}$ on MPO vectors. Returns diagonal coefficients <code>a</code> , off-diagonal <code>b</code> (<code>b[1] = \ seed\ _F</code>), the normalized Krylov basis MPOs, and <code>norm0</code> . Stops early at an invariant subspace.
<code>eval_haydock_cf(a, b, z)</code> $\rightarrow$ <code>ComplexF64</code>	Evaluate the continued fraction $\langle \text{seed} (z - H)^{-1} \text{seed} \rangle$ by backward recursion; <code>b[1]</code> must be <code>norm0</code> .
<code>haydock_resolve_mpo(a, b, basis, z; maxdim=200, cutoff=1e-8)</code> $\rightarrow$ <code>MPO</code>	Reconstruct $(z - H)^{-1} \text{seed} \rangle$ by solving the $N \times N$ Lanczos tridiagonal system $(zI - T)c = b_1 e_1$ and forming $\Pi_0(z) = \sum_n c_n \cdot \text{basis}_n$.
<code>get_bubble_mpo_haydock(H1, H2, ωlist; N_steps=30, η=1e-2, maxdim=200, cutoff=1e-8, ϵ_F=0.0, P_method=:purification, purify_method=:mcweeny, purify_maxdim=40, purify_maxiters=30, purify_tol=1e-5, Ncheb=150, verbose=false)</code> $\rightarrow$ <code>Vector{MPO}</code>	Bare polarization bubble $\Pi_0(\omega)$ as an L -site MPO for each ω in ω list. Builds the Krylov basis once from seed $\Phi_0 = \mathbf{I} \otimes P_2 - P_1 \otimes \mathbf{I}$ on $H_{\text{eff}} = \mathbf{I} \otimes H_2 - H_1 \otimes \mathbf{I}$; for each ω the resolvent $(\omega + i\eta - H_{\text{eff}})^{-1} \Phi_0$ is recovered from the stored Lanczos data. Output compatible with <code>rpa_wynn_from_bubbles</code> .

Double Chebyshev (2D) Bubble Decomposition

Alternative to running KPM directly on the $2L$ -site H_{eff} (where bond dimension grows with each Chebyshev step due to inter-subsystem entanglement). Instead, $G_{\text{eff}}(\omega)$ is decomposed as $G_{\text{eff}}(\omega) \approx \sum_{mn} c_{mn}(\omega) T_m(\tilde{H}_1) \otimes T_n(\tilde{H}_2)$, where T_m, T_n are Chebyshev polynomials of the L -site rescaled Hamiltonians and $c_{mn}(\omega)$ are cheap scalar 2D Chebyshev coefficients (via 2D DCT-II).

The bubble is then $\Pi_0(\omega) = \sum_{mn} c_{mn}(\omega) D_{mn}$ with $D_{mn} = (T_m(\tilde{H}_1)P_1) \odot T_n(\tilde{H}_2) - T_m(\tilde{H}_1) \odot (T_n(\tilde{H}_2)P_2)$ (Hadamard product $\odot$ via `hadamard_mpo`). Plain, Tucker-compressed, and per- ω -SVD variants trade off precomputation cost against the number of Hadamard/QFT operations; all share the keyword arguments listed for `get_bubble_mpo_cheb2d` unless noted otherwise.

Signature	Description
<code>chebyshev2d_gf_coeffs(ω, scale1, center1, scale2, center2, η, N) $\rightarrow$ Matrix{ComplexF64}</code>	2D Chebyshev expansion coefficients c_{mn} of $f(x, y) = 1/(\omega + i\eta - (\text{scale2} \cdot y + \text{center2} - \text{scale1} \cdot x - \text{center1}))$ on $[-1, 1]^2$ via an $N \times N$ Chebyshev-Gauss grid and 2D DCT-II. N should equal <code>length(Tn_list)</code> from <code>KPM_Tn</code> (i.e. <code>Ncheb+1</code>).
<code>get_bubble_mpo_cheb2d(H1, H2, ωlist; Ncheb=50, maxdim=200, cutoff=1e-8, ϵ_F=0.0, P_method=:purification, purify_method=:mcweeny, purify_maxdim=40, purify_maxiters=30, purify_tol=1e-5, η=1e-3, coeff_tol=1e-12, verbose=false) $\rightarrow$ Vector{MPO}</code>	Full-MPO bubble $\Pi_0(\omega)$ for each ω via the plain double Chebyshev decomposition. All $C[m, n](\omega)$ are precomputed once; the (m, n) double loop runs once, each D_{mn} is computed once and accumulated into every $\Pi(\omega)$. <code>coeff_tol</code> prunes negligible (m, n) pairs/rows.
<code>get_bubble_mpo_cheb2d_tucker(H1, H2, ωlist; <get_bubble_mpo_cheb2d kwargs>, tucker_tol=1e-3, tucker_maxrank=20, kernel=:jackson, hooi_iters=3) $\rightarrow$ Vector{MPO}</code>	Tucker-accelerated variant. HOSVD+HOOI finds global bases U_m ($N \times r_m$) and V_n ($N \times r_n$) with $C[m, n](\omega) \approx U_m G(\omega) V_n^\dagger$. Computes $r_m + r_n$ weighted MPO sums and $r_m \times r_n$ Hadamard products once; per- ω cost is $r_m \times r_n$ scalar-weighted MPO additions. Speedup: $N^2 \rightarrow r_m r_n$ Hadamard products. <code>kernel=:jackson</code> applies <code>_jackson_kernel</code> damping to reduce Tucker rank.
<code>get_bubble_diag_cheb2d(H1, H2, ωlist; <get_bubble_mpo_cheb2d kwargs>, qft_tol=1e-9, qft_maxdim=100) $\rightarrow$ Vector{MPS}</code>	Diagonal-only variant of <code>get_bubble_mpo_cheb2d</code> . Returns $\text{diag } \Pi_0(k, \omega) = \langle k \Pi_0(\omega) k \rangle$ as one MPS per ω . Each D_{mn} is QFT-conjugated and diagonal-extracted once; $c_{mn}(\omega) \cdot \text{diag}(D_{mn})$ is accumulated as cheap MPS sums.
<code>get_bubble_diag_cheb2d_svd(H1, H2, ωlist; <get_bubble_diag_cheb2d kwargs>, svd_tol=1e-6, svd_maxrank=20, kernel=:jackson) $\rightarrow$ Vector{MPS}</code>	Per- ω SVD-accelerated variant of <code>get_bubble_diag_cheb2d</code> . Each $C[m, n](\omega)$ is rank-truncated by its own SVD ($C(\omega) = \sum_s \sigma_s u_s v_s^\dagger$, typically rank 2–5 with the Jackson kernel); one Hadamard product and one QFT per (ω, s) .
<code>get_bubble_diag_cheb2d_tucker(H1, H2, ωlist; <get_bubble_diag_cheb2d kwargs>, tucker_tol=1e-3, tucker_maxrank=20, kernel=:jackson, hooi_iters=3) $\rightarrow$ Vector{MPS}</code>	Tucker (HOSVD+HOOI) variant of <code>get_bubble_diag_cheb2d</code> . Global bases U_m, V_n shared across all ω ; $r_m \times r_n$ Hadamard+QFT operations total, then per- ω cost is scalar-weighted MPS additions over the core tensor $\tilde{A}[s_1, s_2, \omega] = U_m^\top C(\omega) V_n$.

Low-Level Bubble Pipeline

Legacy 4-step pipeline retained as building blocks (the high-level `get_bubble_mpo/get_rpa_susceptibility` are preferred for new code):

1. `get_Tnlists` — build Chebyshev lists for system 1, system 2, and the effective combined Hamiltonian $H_{\text{eff}} = \mathbf{I} \otimes H_2 - H_1 \otimes \mathbf{I}$.
2. `get_bubble_expanded_from_Tn` — compute $\Pi_0(\omega)$ as a $2L$ -site MPO.
3. `build_bubble_mpo` — collapse to an L -site MPO.
4. `rpa_from_bubble_diag` — solve the Dyson equation via DMRG linear solver.

Signature	Description
<code>get_Tnlists(H, H2, sites, sites2, N; a=6, maxdim=100) → (Tn1, Tn2, Tn_eff)</code>	Build 3 Chebyshev lists: individual and effective.
<code>get_bubble_expanded_from_Tn(Tn1, Tn2, Tn_eff, s1, s2, N, ω, ε_F; a=6, maxdim=200) → MPO</code>	Lindhard bubble $\Pi_0(\omega)$ as a $2L$ -site MPO.
<code>build_bubble_mpo(ω; Tn1, Tn2, Tn_eff, sites, sites2, finalsites, finalfinalsites, ...) → MPO</code>	Collapse $2L$ -site bubble to L -site MPO.
<code>rpa_from_bubble_diag(Π, V, finalsites, finalfinalsites; nsweeps=20, maxdim=400, cutoff=1e-8) → MPS</code>	Solve $(I - \Pi_0 V)\chi = \Pi_0$ via DMRG linear solver.

Tensor Utilities (used internally by RPA)

Signature	Description
<code>mpo_kron(A, B) → MPO</code>	Concatenate two MPOs (Kronecker product on site space).
<code>mps_kron(A, B) → MPS</code>	Concatenate two MPS.
<code>replace_sites(MPO, newsites) → MPO</code>	Remap physical indices of an MPO to <code>newsites</code> , preserving prime levels.
<code>interleave_mpo(mpo, phys_sites, n) → MPO</code>	Embed L -site MPO into $2L$ -site space by interleaving with identity. <code>n=0</code> : operator at even positions; <code>n=1</code> : operator at odd positions.
<code>interleave_mpo_tb(op, sites_A, sites_B, which) → MPO</code>	Generalization of <code>interleave_mpo</code> for heterogeneous site spaces (Layer, Honeycomb, ...). Embeds <code>op</code> into the $2N$ -site product space ordered <code>[sites_A[1], sites_B[1], ...]</code> . <code>which=:A</code> or <code>:B</code> selects which site set <code>op</code> acts on.
<code>swap_every_other_legs(MPOin, newsites) → MPO</code>	Replace site indices and swap bra $\leftrightarrow$ ket on every even-numbered site; converts a $2L$ -site interleaved bubble MPO into the form expected by <code>collapse_mpo_pairs</code> .
<code>collapse_mpo_pairs(mpo2L, out_sites) → MPO</code>	Merge consecutive site pairs $(2n-1, 2n)$ to single site: $2L \rightarrow L$.
<code>extract_diagonal_to_mps(M) → MPS</code>	Extract diagonal of an MPO as an MPS by projecting bra=ket.

<code>merge_mps_to_mpo(mps) → MPO</code>	Contract consecutive pairs of an MPS into MPO tensors.
<code>convert_mpo(old_mps, new_sites) → MPO</code>	Convert a $2N$ -site MPS (typically from QTCI) into an N -site MPO by merging pairs via <code>merge_mps_to_mpo</code> and remapping physical indices to <code>new_sites</code> .
<code>apply_interleave_swaps(W, sites; cutoff=1e-16, maxdim=200, verbose=false) → MPO</code>	Reorder sites from block order $[1 \dots N, N+1 \dots 2N]$ to interleaved via SWAP gates.

Self-Consistent Mean Field (SCF_tk.jl)

Generic self-consistent mean-field (Hartree–Fock-like) loops for `TBHamiltonian`, built around a density $\rightarrow$ mean-field $\rightarrow$ density fixed-point iteration. The physical channel (CDW, magnetism, s -/ p -wave pairing) is encoded entirely in user-supplied `hartree_builder`/`fock_builder` closures of the form `builder(density, sites) -> MPO`, so all channels share the same generic driver `scf_meanfield`. The density matrix at each iteration comes from `get_density (density_method=:sp2|:mcwee`

Generic Driver

Signature	Description
<code>scf_meanfield(H0::TBHamiltonian, hartree_builder; initial_density=nothing, initial_hartree=nothing, fock_builder=nothing, initial_fock=nothing, density_method=:sp2, density_mode=:direct, Nel=H0.N÷2, fermi=0.0, Ncheb=100, scale=nothing, spectral_bounds=nothing, purification_scale_padding=1.05, max_scf_iter=30, scf_tol=1e-6, mix=0.4, maxdim=100, cutoff=1e-8, purif_maxiter=40, purif_tol=1e-6, stop_on_increase=false, verbose=true) → NamedTuple</code>	Generic mean-field loop. Each iteration: (1) build $H_{\text{MF}} = H_0 + V_{\text{Hartree}}(+V_{\text{Fock}})$; (2) compute the density matrix via <code>get_density</code> ; (3) extract a density profile (<code>density_mode=:direct</code> or <code>:complement</code>); (4) compute the RMS error against the previous profile, linearly mix with weight <code>mix</code> , and rebuild the Hartree/Fock terms via the supplied builders. <code>stop_on_increase=true</code> returns the best-so-far state early if the residual increases. Returns <code>(converged, stopped_by_residual_increase, iterations, history, rms_error, density_mpo, density_mps, hartree_mpo, fock_mpo, ham)</code> , with <code>history</code> a <code>Vector{NamedTuple}</code> of per-iteration diagnostics.

Channel-Specific Drivers

Signature	Description
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<pre> scf_magnetic_hubbard(H0::TBHamiltonian, U::Union{Number,MP0}; initial_up=nothing, initial_dn=nothing, background=0.5, density_method=:sp2, Nel_up=H0.N÷2, Nel_dn=H0.N÷2, fermi=0.0, Ncheb=100, scale=nothing, purification_scale_padding=1.05, max_scf_iter=30, scf_tol=1e-6, mix=0.4, maxdim=100, cutoff=1e-8, purif_maxiter=40, purif_tol=1e-6, verbose=true) → NamedTuple </pre>	Two-channel collinear magnetic (Hubbard) mean field: $H_{\uparrow} = H_{0\uparrow} + U(n_{\downarrow} - \text{background})$, $H_{\downarrow} = H_{0\downarrow} + U(n_{\uparrow} - \text{background})$. If H0 is spinful, $H_{0\uparrow}/H_{0\downarrow}$ are obtained by projecting out the spin core; if spinless both channels share H0. U::MP0 gives a non-local Hartree kernel via <code>hartree_mpo_from_density</code>. Default initial guess is the staggered (Néel) profile from <code>staggered_magnetic_initial</code>. Returns (converged, iterations, rms_error, rho_up, rho_dn, density_up_mpo, density_dn_mpo, H_up, H_dn, history).
<pre> scf_swave_superconducting(H0::TBHamiltonian, g::Number; initial_delta=0.1, initial_up=nothing, initial_dn=nothing, pairing_sign=-1.0, include_hartree=false, hartree_coupling=g, hartree_sign=-1.0, background=0.5, mu=0.0, density_method=:mcweeny, normal_density_mode=:particle, Ncheb=100, scale=nothing, purification_scale_padding=1.05, max_scf_iter=30, scf_tol=1e-6, mix=0.4, maxdim=100, cutoff=1e-8, purif_maxiter=40, purif_tol=1e-6, verbose=true) → NamedTuple </pre>	Spin-singlet <i>s</i>-wave BdG pairing SCF: $\Delta_{\text{new}}(i) = \text{pairing_sign} \cdot g \cdot F_{\text{singlet}}(i)$, with F_{singlet} extracted from the Nambu off-diagonal block of the BdG density matrix via <code>swave_anomalous_profile</code>. <code>include_hartree=true</code> additionally self-consistently updates spin-resolved onsite Hartree potentials $V_{\uparrow/\downarrow}(i) = \text{hartree_sign} \cdot \text{hartree_coupling} \cdot (n_{\downarrow/\uparrow}(i) - \text{background})$ via <code>swave_normal_profiles</code>. If H0 is spinless, a spin degree of freedom is added to a working copy via <code>add_spin!</code> before building the BdG Hamiltonian. Returns (converged, iterations, rms_error, delta_mps, anomalous_mps, rho_up_mps, rho_dn_mps, hartree_up_mpo, hartree_dn_mpo, density_mpo, ham, history).
<pre> scf_swave_hubbard(H0::TBHamiltonian, U::Number; pairing_sign=-1.0, hartree_coupling=U, hartree_sign=-1.0, kwargs...) → NamedTuple </pre>	Attractive on-site Hubbard <i>s</i>-wave SCF; thin wrapper around <code>scf_swave_superconducting</code> with <code>include_hartree=true</code>. Positive U gives attraction: $\Delta_i = -UF_i$, $V_{\uparrow/\downarrow} = -U(n_{\downarrow/\uparrow} - \text{background})$.
<pre> scf_pwave_equalspin(H0::TBHamiltonian, V::Number; initial_up=0.1, initial_dn=nothing, distance=1, eta_down=-1.0, pairing_sign=-1.0, mu=0.0, density_method=:mcweeny, Ncheb=100, scale=nothing, purification_scale_padding=1.05, max_scf_iter=30, scf_tol=1e-6, mix=0.4, maxdim=100, cutoff=1e-8, purif_maxiter=40, purif_tol=1e-6, verbose=true) → NamedTuple </pre>	Minimal spinful equal-spin <i>p</i>-wave triplet SCF on nearest-neighbor (distance) bonds: $\Delta_{\uparrow/\downarrow}(i) = \text{pairing_sign} \cdot V \cdot F_{\uparrow/\downarrow}(i, i + \text{distance})$, with $F_{\uparrow/\downarrow}$ from <code>pwave_equalspin_anomalous_profiles</code>. <code>eta_down=-1</code> (default) seeds a helical configuration ($\Delta_{\downarrow}(i) = -\Delta_{\uparrow}(i)$ initially); <code>eta_down=1</code> starts both channels with the same sign. Returns (converged, iterations, rms_error, delta_up_mps, delta_dn_mps, anomalous_up_mps, anomalous_dn_mps, density_mpo, ham, history).

Hartree/Fock Builder Factories

Each factory returns a closure `builder(density, sites) -> MPO` suitable for `hartree_builder/fock_builder` in `scf_meanfield`, or for the `U/interaction` arguments of `get_scf`.

Signature	Description
<code>hartree_mpo_from_density(density_mps::MPS, interaction_op::MPO, sites; background=nothing, maxdim=100, cutoff=1e-8) -> MPO</code>	Apply interaction kernel <code>interaction_op</code> to $(\rho - \text{background})$ and convert the resulting coefficient MPS to a diagonal Hartree MPO via <code>mps_to_diagonal_mpo</code> . <code>background</code> may be <code>nothing</code> , a <code>Number</code> , or an <code>MPS</code> .
<code>fock_mpo_from_density(density_mpo::MPO, interaction_op::MPO, sites; sign=-1, maxdim=100, cutoff=1e-8) -> MPO</code>	Fock/exchange MPO $F_{ij} = \text{sign} \cdot V_{ij} \cdot \rho_{ij}$ via <code>hadamard_mpo</code> .
<code>fock_exchange_builder(interaction_op::MPO, sign=-1, maxdim=100, cutoff=1e-8) -> Function</code>	Wrap <code>fock_mpo_from_density</code> as a <code>fock_builder(density_mpo, sites)</code> closure for <code>scf_meanfield</code> .
<code>cdw_hartree_builder(U::Number; background=0.5, maxdim=100, cutoff=1e-8) -> Function</code>	Local spinless CDW Hartree builder: $V_H(n) = U(\rho(n) - \text{background})$.
<code>dense_hartree_builder(V, L, sites; background=nothing, type=Float64, tol=1e-8, maxdim=100, cutoff=1e-8, kwargs...) -> Function</code>	Hartree builder for a dense interaction kernel. <code>V</code> is either an existing MPO or a function $V(i, j)$ of 0-indexed coordinates, in which case it is QTCI-compressed via <code>get_mpo</code> .
<code>pair_distance_interaction_mpo(L, sites, distance::Integer, weight=1; type=Float64, tol=1e-8, maxdim=100, cutoff=1e-8) -> MPO</code>	Sparse interaction MPO for fixed-distance site pairs $(i, i+d)$, built from non-cyclic <code>shift_pair_mpos</code> . <code>weight</code> may be a <code>Number</code> , <code>Function</code> $w(i)$, or <code>MPO</code> . The vector form sums multiple <code>distance => weight</code> (or <code>(distance, weight)</code>) terms.
<code>pair_distance_interaction_mpo(L, sites, terms::AbstractVector; kwargs...) -> MPO</code>	
<code>pair_distance_hartree_builder(L, sites, distance_or_terms, weight=1; background=nothing, type=Float64, tol=1e-8, maxdim=100, cutoff=1e-8) -> Function</code>	Hartree builder wrapping <code>pair_distance_interaction_mpo</code> via <code>hartree_mpo_from_density</code> .

Unified Entry Points

Signature	Description
<code>get_scf(H0::TBHamiltonian, channel::Symbol; interaction=:dense, kwargs...) -> NamedTuple</code>	Reads the interaction from <code>H0.interaction_mpo</code> (set via <code>add_interaction!</code>) and dispatches to the 3-argument <code>get_scf(H0, U, channel)</code> . Supported: <code>:cdw</code> , <code>:magnetic</code> . <code>:swave/pwave</code> raise an informative error since they require an explicit coupling constant.

```

get_scf(H0::TBHamiltonian, U,
channel::Symbol; interaction=:local,
background=0.5, distance=1,
initial_density=nothing,
initial_hartree=nothing,
density_mode=:direct,
density_method=:sp2, method=nothing,
Nel=H0.N÷2, fermi=0.0, Ncheb=100,
scale=nothing,
spectral_bounds=nothing,
purification_scale_padding=1.05,
tol=1e-6, maxiters=30, mixing=0.4,
maxdim=100, cutoff=1e-8,
purif_maxiter=40, purif_tol=1e-6,
stop_on_increase=false, verbose=true,
builder_kwargs...) → NamedTuple

```

Convenience front-end dispatching on `channel` (canonicalized via `_canonical_channel`, with aliases). `:CDW` (aliases `:charge`, `:hartree`): builds a Hartree builder according to `interaction` $\in \{ :local, :dense, :distance \}$ — `:local` requires numeric `U` (`cdw_hartree_builder`), `:dense` uses `dense_hartree_builder`, `:distance/:pair(s)` uses `pair_distance_hartree_builder` (vector `U` for multiple terms) — then calls `scf_meanfield`. `:magnetic` (aliases `:magnetism`, `:spin`, `:hubbard`) calls `scf_magnetic_hubbard`. `:swave` (aliases `:s_wave`, `:superconducting`, ...) requires numeric `U` and calls `scf_swave_hubbard`. `:pwave` (aliases `:p_wave`, `:triplet`) requires numeric `U` and calls `scf_pwave_equalspin`. `method/density_method` accept aliases (`:purification`, `:sp2` $\rightarrow$ `:sp2`; `:mcweeny` $\rightarrow$ `:mcweeny`; `:kpm`, `:chebyshev` $\rightarrow$ `:kpm`) via `_canonical_density_method`.

Density and Pairing Profile Extraction

Signature	Description
<code>density_profile_from_dm(density_mpo::MPO, sites; mode=:direct) → MPS</code>	Diagonal of a density-matrix MPO as an MPS. <code>mode=:complement</code> returns $1 - \text{diag}(D)$.
<code>staggered_magnetic_initial(H::TBHamiltonian, amplitude=0.05, background=0.5) → (rho_up, rho_dn)</code>	Antiferromagnetic initial guess for <code>scf_magnetic_hubbard</code> : $\rho_{\uparrow/\downarrow}(n) = \text{background} \pm \text{amplitude} \cdot (-1)^n$. If <code>H</code> is spinful, the spin core is projected out first.
<code>swave_anomalous_profile(density_mpo::MPO, Hbdg::TBHamiltonian; maxdim=100, cutoff=1e-8) → MPS</code>	Local singlet anomalous profile from the Nambu particle-hole block of a spinful <i>s</i> -wave BdG density matrix, projected onto the spin-singlet combination $\frac{1}{2}(f_{\uparrow\downarrow} - f_{\downarrow\uparrow})$. Requires <code>Hbdg.nambu_s</code> and <code>Hbdg.spin_s</code> to be set.
<code>swave_normal_profiles(density_mpo::MPO, Hbdg::TBHamiltonian; mode=:particle, maxdim=100, cutoff=1e-8) → (rho_up, rho_dn)</code>	Spin-resolved local normal densities from a spinful BdG density matrix. <code>mode=:particle</code> reads the particle-particle Nambu block directly; <code>mode=:hole_complement</code> returns $1 - \text{diag}(\text{hole-hole block})$.

`pwave_equalspin_anomalous_profiles(density_mpo, neighbor (distance) equal-spin
Hbdg::TBHamiltonian; distance=1, triplet anomalous bond profiles from the
maxdim=100, cutoff=1e-8) → (F_up, Nambu off-diagonal block of a spinful BdG
F_dn) density matrix; the profile value at i is the
bond $(i, i+distance)$.`

Néel/Trivial Initial-Guess Density Matrices

Seeds for mean-field SCF on interacting models; each returns a diagonal density (MPO, MPS) pair built via QTCI of an explicit occupation function.

Signature	Description
<code>initial_guess_trivial_up_1D(L, sites) → (MPO, MPS)</code>	1D spin-up Néel seed: occupation $x \bmod 2$ on site x .
<code>initial_guess_trivial_down_1D(L, sites) → (MPO, MPS)</code>	1D spin-down Néel seed: occupation $(x+1) \bmod 2$ on site x .
<code>initial_guess_Neel_up(Lx, Ly, sites) → (MPO, MPS)</code>	2D checkerboard spin-up seed: occupation 1 where $i_x + i_y$ is even.
<code>initial_guess_Neel_dn(Lx, Ly, sites) → (MPO, MPS)</code>	2D checkerboard spin-down seed: occupation 1 where $i_x + i_y$ is odd.

Utilities (utils.jl)

Binary Encoding and MPS/MPO Conversions

Signature	Description
<code>to_binary_vector(n, L) → Vector{String}</code>	Convert integer n to length- L big-endian binary string vector.
<code>binary_to_MPS(n, L, sites) → MPS</code>	Computational-basis state $ n\rangle$ as an L -site MPS.
<code>matrix_checker(mpo, sites, i, j) → Number</code>	Matrix element $\langle i \text{mpo} j \rangle$ by contracting basis MPS (small systems only). Optional L argument accepted but ignored.
<code>get_matrix(mpo, sites) → Matrix</code>	Full $D \times D$ dense matrix of <code>mpo</code> (feasible for $D \lesssim 512$). Optional L argument accepted but ignored. Works for any site types.
<code>fix_sites(mpo, sites) → MPO</code>	Replace site indices of an MPO (e.g. from TCI) with target <code>sites</code> .
<code>custom_mpo(mps, new_sites) → MPO</code>	Convert a $2N$ -site MPS (interleaved QTCI) into an N -site MPO.
<code>fused_mpo(mps, new_sites) → MPO</code>	Convert an N -site MPS with dim-4 physical indices (fused grid) into an N -site MPO.
<code>mps2mpo(L, sites, density_mps) → MPO</code>	Promote a diagonal MPS to a diagonal MPO via <code>_asdiagonal</code> .
<code>custom_mps(qtt, sites) → MPS</code>	Replace site indices of a TCI MPS with target <code>sites</code> .
<code>build_shift_mpo(sites, q) → MPO</code>	Cyclic shift MPO: maps $ x\rangle \rightarrow x+q\rangle \pmod{2^L}$ on a binary-encoded chain. Built site-by-site using binary addition with carry; q is an integer offset.

<code>mps_to_diagonal_mpo(mps, sites) → MPO</code>	Convert an MPS to a diagonal MPO on <code>sites</code> by replacing each physical index with a bra-ket pair tied by a delta tensor. Used to embed QTCI-compressed 2D functions (e.g. from <code>build_interaction_op_exciton</code>) into MPO form.
--	--

Profile Evaluation and Spatial Sampling

`eval_mps` evaluates a profile MPS pointwise; `spatial_sampling_plan` is the shared geometry-aware sampler — used by `eval_mps_spatial` below and by `get_ldos_spatial` and its GPU counterparts — that turns `num_x/num_y/ grid/reduce/sublattice` kwargs into a list of sample positions.

Signature	Description
<code>eval_mps(A::MPS, n::Int) → Real</code>	Evaluate a profile MPS <code>A</code> at the 0-indexed basis coordinate <code>n</code> by contracting with the basis state $ n\rangle$, using the big-endian (MSB-first) convention of <code>binary_to_MPS</code> . Not the same convention as <code>_eval_diag_mps</code> , which is LSB-first (QFT momentum ordering); mixing the two bit-reverses the site map.
<code>spatial_sampling_plan(L; Lx=nothing, grid=false, reduce=:point, n_sub=1, num_x=0, num_y=nothing, num_avg=1, x_start=1, x_end=2^L, xwin=nothing, ywin=nothing, x_groups=nothing, box_half=0, sublattice=:auto) → NamedTuple</code>	Geometry-aware real-space sampling plan shared by every spatial sampler. Returns <code>(centers, groups, resolve_sublattice, n_sub, stride_x, stride_y, grid, reduce, a, b)</code> . <code>reduce=:point</code> (default) lays out <code>num_x[×num_y]</code> sample positions/groups, optionally averaged over a <code>box_half</code> neighbourhood — cheap, but can alias features narrower than the sampling grid. <code>reduce=:block</code> partitions the system into <code>num_x×num_y</code> power-of-two blocks ($a = \log_2 \text{num}_x$, $b = \log_2 \text{num}_y$; <code>centers</code> mark block midpoints) whose values are obtained by tracing out the within-block bits (see <code>_eval_block_mps</code>) — gap-free and independent of block size, so it scales to $L_x, L_y \gtrsim 14$. For multi-atom unit cells (<code>n_sub>1</code>), <code>sublattice</code> chooses whether to <code>:resolve</code> (one output column per atom), <code>:average</code> (trace out the sublattice), or decide automatically (<code>:auto</code> : resolve only for atomic-scale <code>:point</code> sampling with unit stride and no box averaging; average otherwise). <code>grid=true</code> (needs <code>Lx</code>) lays centers on a 2D <code>num_x×num_y</code> grid over the 0-indexed window <code>xwin/ywin</code> ; otherwise centers follow a 1D linear sweep of <code>x_start...x_end</code> , or the explicit <code>x_groups</code> .

```
eval_mps_spatial(A::MPS;
  num_x=2^L, num_avg=1,
  x_start=1, x_end=2^L,
  x_groups=nothing, box_half=0,
  Lx=nothing) → (values,
  centers, groups)
```

Spatial sampler for a profile MPS such as an SCF density profile (e.g. `res.rho.up`). Builds a sampling plan via `spatial_sampling_plan` and returns, for each group, the mean of `eval_mps(A, x-1)` over the group's (1-indexed) site indices — the same `num_x/num_avg/x_groups/box_half` API as `get_ldos_spatial`, but using direct MPS evaluation instead of a KPM recursion, so sweeping a large system by sampling-and-averaging is cheap. `Lx` (default $L \div 2$) sets the 2D layout used when `box_half>0`.

`_eval_block_mps(A, ixp, iyp, a, b, Lx, Ly)` is the internal helper behind `reduce=:block`: it returns the sum of `A` over one coarse block by tracing out the within-block position bits (contracted with `[1,1]`) and pinning the kept high-order block bits to the coarse pixel (`ixp, iyp`), on the big-endian site order `[iy_MSB..iy_LSB, ix_MSB..ix_LSB]`.

QTCI Compression and MPS/MPO Builders

Signature	Description
<code>get_mps(L, sites, f; type=Float64, tol=1e-8) → MPS</code>	Compress a scalar profile $f(n)$, $n = 0, \dots, 2^L - 1$, into an MPS via QTCI ($L=1$ is built directly as a 2-element tensor).
<code>get_mpo(L, sites, f; type=Float64, tol=1e-8, kwargs...) → MPO</code>	Compress a function into an MPO via QTCI using 0-indexed coordinates $n = 0, \dots, 2^L - 1$. Dispatches on arity: a 1-argument $f(n)$ builds a diagonal MPO via <code>get_mps + mps2mpo</code> ; a 2-argument $f(i, j)$ builds a full interaction MPO from a 2D QTCI of the interleaved legs via <code>custom_mpo</code> .
<code>constant_mps(sites, value) → MPS</code>	Rank-1 MPS with amplitude <code>value</code> on every computational-basis state, via <code>extract_diagonal_to_mps</code> .
<code>rms_error(a, b) → Float64</code>	RMS distance $\sqrt{\langle (a-b) (a-b) \rangle / 2^L}$ between two MPS over all computational-basis states; used as the SCF convergence metric (the <code>rms_error</code> field returned by <code>scf_meanfield</code>).

Shift and Hadamard Operators

Signature	Description
<code>shift_mpo(sites, q; cyclic=false) → MPO</code>	Translation operator $(Tf)(x) = f(x + q)$ on the binary-encoded chain. For $q \geq 0$ built directly via <code>build_shift_mpo</code> ; for $q < 0$ returns the adjoint of <code>build_shift_mpo(sites, -q, cyclic)</code> .
<code>shift_pair_mpos(sites, q; cyclic=false) → (MPO, MPO)</code>	$(K, K^\dagger)$ where $K = \text{shift_mpo}(\text{sites}, q; \text{cyclic})$ — the shift operator and its adjoint, as used by <code>pair_distance_interaction_mpo</code> .

<code>shift_hopping_mpo(hopping, sites, q; cyclic=false, maxdim=typemax(Int), cutoff=1e-12, apply_kwargs=NamedTuple()) → MPO</code>	Symmetrized shifted-hopping term $K \cdot \text{hopping} + K^\dagger \cdot \text{hopping}^\dagger$ for a bond at displacement q , via <code>shift_pair_mpos</code> .
<code>hadamard_mpo(A, B, out_sites; maxdim=100, cutoff=1e-8) → MPO</code>	Site-wise (Hadamard) product $C_{ij} = A_{ij}B_{ij}$ of two MPOs with matching site count, remapped onto <code>out_sites</code> . Used by <code>fock_mpo_from_density</code> to form $F_{ij} = \text{sign} \cdot V_{ij}\rho_{ij}$.

Custom Operator Extensions

Defined in `utils.jl` and `2D_lattice.jl`:

Name	Site type	Matrix
<code>sigma_plus</code>	Qubit	$\begin{pmatrix} 0 & 1 \\ 0 & 0 \end{pmatrix}$ — binary increment (up-shift)
<code>sigma_minus</code>	Qubit	$\begin{pmatrix} 0 & 0 \\ 1 & 0 \end{pmatrix}$ — binary decrement (down-shift)
<code>sigma_d</code>	Qubit	$\begin{pmatrix} 0 & 0 \\ 0 & 1 \end{pmatrix}$ — projector onto $ 1\rangle$
<code>sigma_u</code>	Qubit	$\begin{pmatrix} 1 & 0 \\ 0 & 0 \end{pmatrix}$ — projector onto $ 0\rangle$

Built-in spin- $\frac{1}{2}$ operators (used by `prepend_spin/postpend_spin`):

Symbol	Matrix	Symbol	Matrix
<code>:Id</code>	$\mathbf{I}_2$	<code>:Sz</code>	$\frac{1}{2}\sigma_z = \text{diag}(\frac{1}{2}, -\frac{1}{2})$
<code>:Pup</code>	$\text{diag}(1, 0)$	<code>:Sp</code>	$ \uparrow\rangle\langle\downarrow $
<code>:Pdn</code>	$\text{diag}(0, 1)$	<code>:Sm</code>	$ \downarrow\rangle\langle\uparrow $
<code>:Sx</code>	$\frac{1}{2}\sigma_x$	<code>:Sy</code>	$\frac{1}{2}\sigma_y$
<code>:iSy</code>	$i\sigma_y = \begin{bmatrix} 0 & 1 \\ -1 & 0 \end{bmatrix}$	<code>:miSy</code>	$-i\sigma_y$

Built-in Nambu operators (used by `prepend_nambu/postpend_nambu`):

<code>:Id</code>	$\mathbf{I}_2$	<code>:tz</code>	$\tau_z = \text{diag}(1, -1)$
<code>:Pp</code>	$\text{diag}(1, 0)$	<code>:tp</code>	$ p\rangle\langle h $
<code>:Ph</code>	$\text{diag}(0, 1)$	<code>:tm</code>	$ h\rangle\langle p $
<code>:tx</code>	σ_x	<code>:ty</code>	σ_y

Quick-Reference Workflows

KPM Density of States and Band Structure

```
# 1. Build Hamiltonian (scale=0 means lazy DMRG estimation)
H = get_Hamiltonian("square_2d", 1.0; L=8)
```

```
# 2. Build Chebyshev list (caches in H._tn_cache / H.scale / H.center)
Tn, scale, center = KPM_Tn(H, 200; maxdim=100)
```

```
# 3. Sweep in energy (always rescale before calling KPM functions)
omega_r = (omega_phys - center) / scale    # must be in (-1, 1)

# 4. Observables
A = get_ldos_w_from_Tn(Tn, 200, omega_r; maxdim=100) # delta(omega-H)
dos  = real(tr(A))                                # density of states
G    = get_Green_retarded_from_Tn(Tn, 200, omega_r; eta=0.02)
rho  = get_density_from_Tn(Tn, 200; fermi=omega_r_fermi)

# 5. Band structure A(k,omega) -- works for plain, spinful, BdG, layered H
Akw = get_bands(H, omega_phys_vals)          # trace over aux DOF
Akw = get_bands(H, omega_phys_vals; aux_proj=[1]) # spin-up only
```

Density Matrix Purification

```
# McWeeny: one-liner (initial guess built and cached internally)
rho_mc = mcweeny_purify(H; maxiters=30, maxdim=200, tol=1e-5, verbose=true)

# SP2: one-liner with explicit electron count
rho_sp = sp2_purify(H; Nel=H.N div 2, maxiters=40, maxdim=200)

# Low-level form (for non-standard center or external MPO):
rho0 = purification_initial_guess(H; maxdim=100, cutoff=1e-8)
rho  = mcweeny_purify(rho0; maxiters=30, maxdim=200)
```

Topological Invariants

```
# Define coordinate functions (0-indexed site i, 0-indexed length L_chain)
xfunc(i, L) = Float64(i % Int(sqrt(L))) # x-coordinate on square lattice
yfunc(i, L) = Float64(i div Int(sqrt(L))) # y-coordinate
sz_func(i)  = isodd(i) ? 1.0 : -1.0      # sublattice / sigma_z (1-indexed)

# 1D winding number (returns a closure; pre-computes 2 MPOs once)
winding = get_W(H, xfunc, sz_func; quenched=true, Lambda=10.0, maxdim=40)
nu = sum(real(winding(alpha)) for alpha in 1:H.N) / H.N

# 2D Chern marker (returns a closure; pre-computes C1--C4 MPOs once)
chern = get_C(H, xfunc, yfunc; quenched=true, Lambda=10.0, maxdim=500)
C_local = real(chern(H.N div 2)) # evaluate at central site
C_avg   = sum(real(chern(a)) for a in 1:H.N) / H.N
```

Bilayer and Twisted Multilayer

```
# Bernal bilayer graphene (exact interlayer, no TCI)
H_bg = bilayer_hamiltonian(:honeycomb, 4, 3;
    stacking=:Bernal, t_intra=1.0, t_inter=0.3)
# H_bg.sites = [layer_s, pos_qubits...]

# Twisted bilayer at 5 degrees
H_tbg = twisted_bilayer_hamiltonian(:triangular, 4, 4, 5.0;
    t_intra=1.0, t_inter=0.3, alpha_decay=1/16)
```

Spinful Superconductor (BdG)

```
H = get_Hamiltonian("square_2d", 1.0; L=8)
add_onsite!(H, _ -> -2.0)           # chemical potential
add_zeeman!(H, 0.1; direction=:z)   # spin-splitting
add_soc!(H, 0.05; type=:rashba)     # Rashba SOC
add_superconductivity!(H, 0.3)      # singlet pairing
# H.sites = [nambu_s, spin_s, pos_qubits...] (aux_side=:pre)
# H.N is unchanged; the MPO now spans 2*2*N = 4N states

# Band structure tracing over Nambu and spin:
Akw = get_bands(H, omega_phys_vals)
# Band structure in the particle sector only:
Akw = get_bands(H, omega_phys_vals; aux_proj=[1, 1])
```

Band Structure with Sublattice Resolution

```
# Sublattice-aware honeycomb (H.sublattice_s is set)
H = get_Hamiltonian("honeycomb", (t=1.0,); L=8, Lx=4, Ly=4)
KPM_Tn(H, 200) # not needed -- get_bands runs online
Ak = get_bands(H, 200, 2, omega_vals) # sum over sublattices
Ak_A = get_bands(H, 200, 2, omega_vals;
                  sublat_proj=true, proj_sl=1) # sublattice A only

# High-symmetry path (G--M--K'--G for honeycomb)
res = get_bands(H, 200, 2, omega;
                kpath=[:G, :M, :Kp, :G], kpath_lattice=:honeycomb, num_x=40)
heatmap(1:size(res.Ak,2), omega, res.Ak; xticks=(res.ticks, res.labels))

# Legacy sublattice projection (models without H.sublattice_s, e.g. "hex_2d")
H2 = get_Hamiltonian("hex_2d", 1.0; L=8)
Ak = get_bands(H2, 200, 2, omega_vals; sublattice=true, proj_sl=1)
```

aux_side: postpending auxiliary DOF

By default (aux_side=:pre) auxiliary indices are prepended. To instead append them, pass position=:post to the first extension function:

```
H = get_Hamiltonian("square_2d", 1.0; L=8)
add_spin!(H; position=:post) # H.sites = [pos_qubits..., spin_s]
add_superconductivity!(H, 0.1) # H.sites = [pos_qubits..., spin_s, nambu_s]
# All subsequent add_* calls inherit H.aux_side = :post
```

Krylov Green's Function (krylov_tk.jl)

Computes the retarded single-particle Green's function $G(\omega) = (\omega + i\eta - H)^{-1}$ as an MPO by solving the vectorized linear system $[(\omega + i\eta - H) \otimes I] |G\rangle\rangle = |I\rangle\rangle$ via `ITensorMPS.linsolve` (a DMRG-like Krylov solver). No KPM Chebyshev expansion is required; the unscaled Hamiltonian is used directly.

Signature

Description

<code>get_green_krylov(H_mpo, sites, ω_phys; η=1e-2, nsweeps=12, maxdim=100, cutoff=1e-8, x0_mpo=nothing, ishermitian=false, tol=1e-10, maxiter=600, krylovdim=30, verbose=false) → MPO</code>	Low-level form. Solve on raw MPO <code>H_mpo</code> .
<code>get_green_krylov(H::TBHamiltonian, ω_phys; ...) → MPO</code>	High-level form. Forwards to the MPO overload using <code>H.mpo</code> and <code>H.sites</code> .

Key kwargs:

- `x0_mpo`: optional MPO warm-start (e.g. a cheap KPM Green's function). Vectorized via `_vec_mps_from_mpo` and passed as initial guess to `linsolve`.
- `ishermiteian`: set `true` only when the shifted operator is truly Hermitian (not physical for retarded GF with real $\eta > 0$).
- `krylovdim`: Krylov subspace dimension per site (default 30).

Usage:

```
G = get_green_krylov(H, omega; eta=0.05, nsweeps=20, maxdim=200)
dos = -imag(tr(G)) / pi
ldos = real(inner(psi_i, apply(G, psi_i))) # LDoS at site i
```

```
# Warm-start from KPM estimate
```

```
G_kpm = get_Green_retarded_from_Tn(H._tn_cache, H._tn_Ncheb, (omega-H.center)/H.scale; eta)
G      = get_green_krylov(H, omega; x0_mpo=G_kpm, nsweeps=8, maxdim=300)
```

Flake Masking (Flake_tk.jl)

Restricts a Hamiltonian to an irregular geometry by applying a smooth diagonal mask $M_i = \sigma(\text{sdf}(\mathbf{r}_i)/\sigma_0)$, where $\text{sdf}(\mathbf{r})$ is a signed-distance function (positive inside the flake) and σ is the logistic sigmoid. The resulting $H_{\text{flake}} = MHM$. Call `mask_hamiltonian` *before* `add_spin!`, `add_superconductivity!`, or any bilayer constructor.

SDF Primitives

Signature	Description
<code>sdf_interval(x_lo, x_hi) → x → Float64</code>	1D interval $[x_{\text{lo}}, x_{\text{hi}}]$; positive inside.
<code>sdf_disk(cx, cy, r) → (x,y) → Float64</code>	Disk of radius <code>r</code> centred at <code>(cx,cy)</code> ; positive inside.
<code>sdf_rect(cx, cy, w, h) → (x,y) → Float64</code>	Axis-aligned rectangle of full width <code>w</code> , height <code>h</code> ; positive inside.
<code>sdf_halfplane(nx, ny, d) → (x,y) → Float64</code>	Half-plane $n_x x + n_y y \geq d$; <code>(nx,ny)</code> need not be unit-length.
<code>sdf_annulus(cx, cy, r_in, r_out) → (x,y) → Float64</code>	Circular annulus; implemented as <code>sdf_subtract(disk_out, disk_in)</code> .
<code>sdf_convex_polygon(vertices) → (x,y) → Float64</code>	Convex polygon from CCW vertex list of <code>(x,y)</code> tuples.

CSG Combinators

Function	Description
<code>sdf_union(f, g)</code>	Boolean union: $\max(f, g)$; positive where either region holds.
<code>sdf_intersect(f, g)</code>	Boolean intersection: $\min(f, g)$; positive inside both.
<code>sdf_subtract(f, g)</code>	Subtraction: $\min(f, -g)$; removes g from f .

Mask Application

Signature	Description
<code>mask_hamiltonian(H, sdf; sigma=0.3, tol=1e-8, maxdim=200, cutoff=1e-8) → TBHamiltonian</code>	Apply smooth flake mask to H : learns M_i via QTCI, returns $H_{\text{flake}} = M H M$ as a new TBHamiltonian. sigma : boundary half-width in lattice units; smaller sigma = sharper edge.

Examples:

```
H = get_Hamiltonian("square_2d", 1.0; L=8, Lx=4, Ly=4)

# Disk
H_disk = mask_hamiltonian(H, sdf_disk(8.0, 8.0, 6.0); sigma=0.3)

# Triangular flake (intersection of three half-planes)
tri = sdf_intersect(sdf_intersect(
    sdf_halfplane(0, 1, 2.0),
    sdf_halfplane(-sqrt(3)/2, -0.5, -8.0)),
    sdf_halfplane(sqrt(3)/2, -0.5, -8.0))
H_tri = mask_hamiltonian(H, tri; sigma=0.2)

# Ring
H_ring = mask_hamiltonian(H, sdf_annulus(8.0, 8.0, 3.0, 7.0); sigma=0.3)
```

GPU Acceleration (GPU.tk.jl)

This file is the GPU companion to the CPU solvers in `src/solvers/` and `src/core/Utils.jl`. It makes large production runs (big L , many Chebyshev moments, fine spatial/ k grids) tractable by moving the dominant Chebyshev-recurrence and MPO-product cost onto a GPU via the NDTensors CUDA backend. Every public function mirrors a CPU counterpart with a `_gpu` suffix and (unless its own docstring says otherwise) accepts the same keyword arguments and returns the same shape of result.

Requirements:

```
using CUDA          # must be loaded *before* calling any *_gpu function
include("TensorBinding.jl"); using .TensorBinding
```

```
A = TensorBinding.get_ldos_spatial_gpu(H, 200, omega; maxdim=200)
```

GPU/CPU split (general pattern):

- **GPU:** Chebyshev recurrence (`KPM_Tn_gpu` / inline recursions), weighted MPO sums (`_weighted_mpo_sum_gpu`), MPO×MPO and MPO×MPS products and truncation, Hadamard products (`_hadamard_mpo_gpu`), aux-index projections (`_project_aux_gpu`), diagonal extraction (`extract_diagonal_to_mps_gpu`), and real-space/QFT sampling.
- **CPU:** one-time setup (Hamiltonian/operator construction, Tucker SVDs, k -/spatial-group bookkeeping, KPM weight matrices, McWeeny initial guesses) and the final per- ω scalar accumulation.

Precision (why F32): The NDTensors GPU backend requires Float32 storage, so every MPO/MPS moved to GPU via `_to_gpu_mpo/_to_gpu_mps` is first cast to `ComplexF32`; results moved back via `_to_cpu_mpo/_to_cpu_mps` are promoted to `ComplexF64`. This is fine for the observables computed here, but `ComplexF32` eigendecomposition can produce NaN at very tight `cutoff` on large systems — affected functions warn (without altering the value) if `cutoff` is below a recommended floor, typically $1e-4$ to $1e-6$ depending on the routine. Per the project convention, these warnings never silently clamp `cutoff`; the user’s value is always passed through unchanged.

Real-space / bit-ordering conventions: Real-space sampling (`_eval_mps_bigendian_gpu`, `_eval_block_mps_gpu`, used by `get_ldos_spatial_gpu` and `get_scf_magnetization_gpu`) encodes the position index MSB-first across the site list, matching the CPU `eval_mps/binary_to_MPS` convention exactly. The QFT/bands path (`_eval_diag_mps_gpu`, used by `get_bands_gpu`) instead uses the legacy LSB-first convention required by the quantics-Fourier MPO. The two are **not** interchangeable — see `spatial_sampling_plan` in `Utils.jl` for how the shared sampler keeps them straight.

Spectral and Spatial Maps (DOS/LDOS/Bands)

Signature	Description
<code>get_bands_gpu(H::TBHamiltonian, Ncheb::Int, $\omega_{\text{phys_vals}}$; kwargs...) → Matrix{Float64} or NamedTuple(Ak, ticks, labels)</code>	GPU-accelerated $A(k, \omega)$ bands; mirrors the <code>TBHamiltonian</code> overload of <code>get_bands</code> . GPU handles the full Chebyshev MPO recurrence and the QFT sandwich applied to each Chebyshev moment; CPU handles k -group setup, the KPM weight matrix, and final scalar accumulation. All kwargs (<code>kpath</code> , <code>kpath_lattice</code> , <code>spin_proj</code> , <code>nambu_proj</code> , <code>layer_proj</code> , <code>sublat_proj</code> , <code>num_x</code> , <code>num_y</code> , <code>kernel</code> , <code>maxdim</code> , <code>cutoff</code> , ...) are identical to <code>get_bands</code> . Returns a plain <code>Matrix{Float64}</code> when no <code>kpath</code> is given, or <code>(Ak, ticks, labels)</code> for a high-symmetry path. If <code>Ak</code> comes back all-NaN, raise <code>cutoff</code> rather than lowering it.


```
get_ldos_spatial_gpu(H::TBHamiltonian,
Ncheb::Int,  $\omega$ _phys_vals;
x_groups=nothing, num_x=H.N,
num_y=nothing, num_avg=1, x_start=1,
x_end=H.N, grid=false, xwin=nothing,
ywin=nothing, box_half=0,
reduce=:point, sublattice=:auto,
kernel=:jackson, lambda=4.0,
maxdim=100, cutoff=1e-8,
⟨aux-projection kwargs⟩,
verbose=false, printinfo=false) →
Matrix{Float64} (shape  $N_\omega \times n_{\text{cols}}$ )
```

```
get_dos_stochastic_gpu(H::TBHamiltonian, GPU-accelerated stochastic DOS via MPS
Ncheb::Int,  $\omega$ _phys_vals; N_sample=50,
N_bound=0, seed=42, normalize=false,
dos_weighting=:trace,
kernel=:jackson, lambda=4.0, eta=0.0,
m_order=4, maxdim=100, cutoff=1e-8,
continuum_only=false, ⟨aux-projection
kwargs⟩, verbose=false,
printinfo=false) → Vector{Float64}
(length  $N_\omega$ )
```

GPU-accelerated version of `get_ldos_spatial` (MPO mode only; `:mps` mode is not available on GPU). `reduce=:point` (default) reads the LDOS at `num_x[ $\times$ num_y]` cells / `x_groups`, optionally box-averaged — coarse grids alias thin features. `reduce=:block` integrates over `num_x $\times$ num_y` power-of-two blocks by tracing out the within-block bits (gap-free, the scalable tool for large-scale edge-state maps); columns are row-major over coarse pixels, `col = ixp + iyp $\cdot$ num_x + 1`.

Column layout: no sublattice DOF $\rightarrow (N_\omega \times n_g)$, one column per pixel; sublattice resolved $\rightarrow (N_\omega \times n_g \times n_{\text{sub}})$, interleaved $[A_0, B_0, A_1, B_1, \dots]$; sublattice averaged (large scale / `:block`) $\rightarrow (N_\omega \times n_g)$. GPU handles the entire Chebyshev MPO recurrence, aux projections, diagonal extraction, and real-space scalar sampling; CPU handles the KPM weight matrix and output accumulation.

GPU-accelerated stochastic DOS via MPS Chebyshev KPM. For each random sample the scaled Hamiltonian MPO lives on GPU and the product-state MPS is transferred once before the recursion; the Chebyshev moments $\langle \psi_0 | T_n(\tilde{H}) | \psi_0 \rangle$ are scalars pulled to CPU at each step. `N_bound>0` (exciton bound-sector enrichment) and `dos_weighting=:sample` (unweighted sampled signal `avg_full + avg_bound`, useful for visualising exciton peaks hidden by continuum phase-space factors) are supported. For exciton Hamiltonians, `continuum_only=true` samples ordered electron-hole product states with $x_e \neq x_h$. `kernel=:hodc` selects the Higher-Order Delta Chebyshev reconstruction (`eta`, `m_order` control the contour); its weights already carry the full KPM normalisation, so no $\sqrt{1 - \omega^2}$ denominator is applied (`eta=0` falls back to $1/(N_{\text{cheb}} + 1)$). Otherwise `kernel` is a convolution kernel (`:jackson` default, `:lorentz` with `lambda`, `:fejer`, `:dirichlet`).

```
get_exciton_ldos_spatial_gpu(H::TBHamiltonian,
Ncheb::Int, ω_phys_vals;
X_list=nothing, X_groups=nothing,
x_groups=nothing, num_x=H.N,
num_avg=1, x_start=1, x_end=H.N,
kernel=:jackson, lambda=4.0, eta=0.0,
m_order=4, maxdim=100, cutoff=1e-8,
verbose=false, printinfo=false) →
Matrix{Float64} (shape  $N_\omega \times n_X$ )
```

GPU-accelerated spatial exciton LDOS. For each bound exciton position X (electron = hole = X , 1-indexed in $1:H.N$), the local spectral weight $A(X, \omega) = \langle X, X | \delta(\omega - H) | X, X \rangle$ is reconstructed from the Chebyshev moments $\mu_n = \langle X, X | T_n(\tilde{H}) | X, X \rangle$, with one GPU MPS Chebyshev recursion run per X starting from $|X, X\rangle = \text{mpsexciton}(X, H.\text{sites})$; moments are scalars pulled to CPU. Spatial / batched GPU analog of the CPU `get_exciton_ldos` (same KPM-weight-matrix reconstruction). `X_list` selects positions directly; `X_groups` / `x_groups` select explicit position groups and average all probes in each group into one output column; otherwise `num_x` coarse groups are built over `x_start:x_end` with the same stride convention as `get_ldos_spatial`. `kernel=:hodc` behaves as in `get_dos_stochastic_gpu`.

Topology

Signature	Description
<pre>get_C_gpu(H::TBHamiltonian, xfunc=nothing, yfunc=nothing; method=:mcweeny, fermi=0.0, l=nothing, Λ=10, Lambda=nothing, Nchebychev=300, maxdim=500, cutoff=1e-8, Nel=nothing, quenched=true, printinfo=false) → Function</pre>	GPU-accelerated real-space Chern marker. Mirrors <code>get_C</code> exactly but runs all MPO×MPO products (projector assembly and C_1–C_4 construction) on GPU in F32, and returns the same closure <code>C_at(uc::Int) → ComplexF64</code> as <code>get_C</code>. Differences from <code>get_C</code>: all <code>apply/truncate!</code> run on GPU tensors (F32); a warning is emitted if <code>cutoff < 1e-6</code> since <code>ComplexF32</code> eigendecompositions can NaN at tight cutoffs on large systems; the projector is built on CPU via <code>purification_initial_guess</code> (<code>method=:KPM</code> via <code>_get_projector</code>) and then moved to GPU — for <code>method=:mcweeny/sp2</code> the purification loop itself runs on GPU; <code>sequential</code> mode is not supported (non-sequential <code>quenched</code> is always used). All other keyword arguments are identical to <code>get_C</code>.

SCF (Magnetic Hubbard)

Signature	Description
-----------	-------------

```
scf_magnetic_hubbard_gpu(H0::TBHamiltonian,
U::Union{Number,MPO};
initial_up=nothing,
initial_dn=nothing, background=0.5,
Nel_up=H0.N÷2, Nel_dn=H0.N÷2,
fermi=0.0, scale=H0.scale,
purification_scale_padding=1.05,
max_scf_iter=30, scf_tol=1e-6,
mix=0.4, maxdim=100, cutoff=1e-8,
purif_maxiter=40, purif_tol=1e-6,
verbose=true) → NamedTuple
```

```
get_scf_magnetization_gpu(res;
num_x=0, num_y=nothing, num_avg=1,
x_start=1, x_end=N, x_groups=nothing,
box_half=0, Lx=nothing) → (values,
centers, groups, n_up, n_dn)
```

```
get_scf_bands_gpu(res, Ncheb::Int,
omega; kwargs...) → (Ak, omega,
ticks, labels)
```

GPU-accelerated two-channel collinear magnetic mean-field loop for the on-site Hubbard model:

$$H_{\uparrow} = H_{0\uparrow} + U \cdot \text{diag}(n_{\downarrow} - \text{background}),$$

$$H_{\downarrow} = H_{0\downarrow} + U \cdot \text{diag}(n_{\uparrow} - \text{background}).$$

The SCF iteration keeps density profiles, Hartree MPOs, Hamiltonian MPOs, density matrices, RMS checks, and mixing on GPU; CPU objects are built only for initialization and for the returned compatibility fields. Only `density_method=:mcweeny` is supported (grand-canonical at `fermi`); for particle-number-fixed SP2 use the CPU `scf_magnetic_hubbard`. A concrete nonzero `scale` is **required** so the GPU initial guess can be formed without estimating spectral bounds on CPU during the loop. A warning (not a clamp) is emitted if `cutoff < 1e-5`. Returns (`converged`, `iterations`, `rms_error`, `rho_up`, `rho_dn`, `density_up_mpo`, `density_dn_mpo`, `H_up`, `H_dn`, `rho_up_gpu`, `rho_dn_gpu`, `density_up_mpo_gpu`, `density_dn_mpo_gpu`, `H_up_mpo_gpu`, `H_dn_mpo_gpu`, `history`) — the `_gpu` fields retain GPU-resident objects for reuse by `get_scf_magnetization_gpu/get_scf_bands_gpu`. Sample the converged magnetic SCF density matrices on GPU and extract only the final scalar values. If `res` carries GPU density MPOs from `scf_magnetic_hubbard_gpu` (`density_up/dn_mpo_gpu`), they are reused directly; otherwise the CPU density MPOs are uploaded once. Sampling groups come from the shared `spatial_sampling_plan` (via the internal `_tb_spatial_groups_gpu` wrapper), and each sampled point is evaluated in the same big-endian real-space convention as `binary_to_MPS`. `values = (n↑ - n↓)/2` is the local magnetization $\langle S_z \rangle(r)$. Compute spin-summed mean-field bands from a converged magnetic SCF result. Deliberately separate from `scf_magnetic_hubbard_gpu`: it initializes from the CPU `res.H_up/res.H_dn`, and each `get_bands_gpu` call uploads once and keeps the Chebyshev/QFT accumulation on GPU, extracting only scalars. `Ak = Ak_up + Ak_dn`; `kwargs` are forwarded verbatim to both `get_bands_gpu` calls.

Internal helpers (not part of the public API): `KPM_Tn_gpu` runs the GPU Chebyshev recurrence and returns the moment MPOs (optionally restricted to `keep_indices`); `extract_diagonal_to_mps_gpu` and `density_profile_from_dm_gpu` are GPU-resident analogs of `extract_diagonal_to_mps` and `density_profile_from_dm` (`mode=:complement` returns $1 - \text{diag}(D)$ on GPU); `_to_gpu_mpo/_to_gpu_mps/_to_cpu_mpo/_to_cpu_mps` handle the F32 round-trip conversions; `_hadamard_mpo_gpu`, `_apply_qft_conj_gpu`, `_project_aux_gpu`, and `_weighted_mpo_sum_gpu` are the GPU primitives used throughout the spectral/spatial map and topology routines; `_check_gpu` verifies CUDA is loaded and `_gpu_gc!` runs incremental GPU garbage collection between steps to bound memory growth in long recursions.

T-Junction Geometries (TJunction.tk.jl)

Builds T/Y-junction Hamiltonians via a spin-1 (dim-3) chain-label index: a T-junction (Y-junction) connects three identical 1D chains at a common site. The dim-3 "TJunction" index postpended to the MPO labels the branch ($1 \equiv |-1\rangle$, $2 \equiv |0\rangle$, $3 \equiv |+1\rangle$); all three branches share the same intra-chain Hamiltonian, and a 3×3 operator acting at a designated junction site couples them.

Atom index convention for `H.geometry` (matches kagome — branch-fast): for combined 1-indexed atom i , $n = (i - 1) \div 3$ is the 0-indexed chain position and $s = (i - 1) \bmod 3 + 1$ is the branch ($1 = |-1\rangle$, $2 = |0\rangle$, $3 = |+1\rangle$); total atoms $3N$ with $N = 2^L$.

TJunction Index and Geometry

Signature	Description
<code>tjunction_index() → Index</code>	Dim-3 Index tagged "TJunction". States: $1 = \text{branch } -1\rangle$, $2 = \text{branch } 0\rangle$, $3 = \text{branch } +1\rangle$.
<code>tjunction_positions(N::Int, junction_site::Int) → Matrix{Float64}</code>	$(3N \times 2)$ real-space position matrix for a Y-junction of three chains of length N , meeting at <code>junction_site</code> (0-indexed). Branches radiate symmetrically from the origin at 0, 120, 240; position along a branch is the signed distance $n - \text{junction_site}$ (positive = away from junction).

Hamiltonian Construction

Signature	Description
-----------	-------------

<pre>add_tjunction!(H::TBHamiltonian, t_j; junction_site=0, coupling=nothing, cutoff=1e-8, maxdim=200) → TBHamiltonian</pre>	Extend a 1D TBHamiltonian in place to a T/Y-junction by postpending a dim-3 "TJunction" index: $H_{\text{TJ}} = I_3 \otimes H_{\text{chain}} + J(t_j) \otimes \text{junction_site}\rangle\langle\text{junction_site}$. The default 3×3 coupling is the symmetric Y-junction $J = t_j \begin{pmatrix} 0 & 1 & 1 \\ 1 & 0 & 1 \\ 1 & 1 & 0 \end{pmatrix}$; pass a custom 3×3 matrix via <code>coupling</code> (e.g. for a linear T-junction where only two of the three branches are mutually coupled). Requires H to have no existing auxiliary indices (spin, Nambu, layer, sublattice). Sets <code>H.geometry</code> to the 2D Y-junction layout (<code>tjunction_positions</code>), <code>H.sublattice_s</code> to the TJunction index, and <code>H.aux_side=:post</code>; invalidates all caches. Standalone constructor: three identical 1D nearest-neighbour chains (hopping <code>t</code>, 2^L sites per branch, boundary BC) meeting at <code>junction_site</code> with coupling <code>t_j</code>. Built via <code>get_Hamiltonian("chain_1d", t; L, boundary)</code> followed by <code>add_tjunction!</code>. Sets a rough spectral scale $2.5 t + 2 \max(J)$.
<pre>tjunction_hamiltonian(L::Int, t::Number, t_j::Number; junction_site=0, coupling=nothing, boundary=:open, cutoff=1e-8, maxdim=200) → TBHamiltonian</pre>	
<pre>tjunction_lattice_hamiltonian(Lx::Int, Ly::Int, L::Int, t::Number, t_j::Number, t_inter::Number; junction_site=0, coupling=nothing, boundary=:open, cutoff=1e-8, maxdim=200) → TBHamiltonian</pre>	Triangular lattice of T/Y-junctions with $2^{L_x} \times 2^{L_y}$ unit cells, each a T-junction with three arms of 2^L sites. Site order is [iy bits(L_y), ix bits(L_x), chain(L), tj_s] (each block MSB-first). Structure: (1) intra-arm NN hopping <code>t</code> along each chain; (2) junction coupling $t_j \begin{pmatrix} 0 & 1 & 1 \\ 1 & 0 & 1 \\ 1 & 1 & 0 \end{pmatrix}$ (or <code>coupling</code>) at <code>junction_site</code>, acting on the branch index; (3) inter-cell hopping <code>t_inter</code> between arm ends across unit cells: an x-shift (-1) couples arm 1 (330) $\leftrightarrow$ arm 3 (210), a y-shift ($+N_x$) couples arm 3 $\leftrightarrow$ arm 2 (90), and a diagonal shift ($+N_x-1$) couples arm 2 $\leftrightarrow$ arm 1. <code>H.sublattice_s</code> stores the dim-3 TJunction index (<code>H.aux_side=:post</code>); <code>H.L=Lx+Ly+L</code>, <code>H.N=2^(Lx+Ly)·2^L</code>. Spectral scale $2.5 t + 2 \max(J) + 2 t_{\text{inter}}$.

Usage:

```
# Symmetric Y-junction, junction at left end (site 0)
H = tjunction_hamiltonian(8, 1.0, 0.5)

# Linear T-junction (branches 1<->2 and 2<->3 only)
H3 = tjunction_hamiltonian(8, 1.0, 0.5; coupling=[0 1 0; 1 0 1; 0 1 0])

# 4x4 triangular lattice of T-junctions, arms of length 8
```

```
Hlat = tjunction_lattice_hamiltonian(2, 2, 3, 1.0, 0.5, 0.8)
```

The internal helper `_site_projector_mpo(L, pos_sites, n)` builds the exact rank-1 (bond-dim-1) projector $|n\rangle\langle n|$ for 0-indexed site `n` on `L` position qubits, via `OpSum` with "`sigma_u`"/"`sigma_d`" factors — no QTCI required.

Non-Hermitian Extensions (NH_tk.jl)

`NH_tk.jl` adds non-Hermitian (NH) Hamiltonian support on top of `TBHamiltonian`. The core construction hermitizes a non-Hermitian single-particle MPO by adding one dim-2 auxiliary block index:

$$H_{\text{NH}}(z) = \begin{pmatrix} 0 & zI - H \\ (zI - H)^\dagger & 0 \end{pmatrix} = |1\rangle\langle 2| \otimes (zI - H) + |2\rangle\langle 1| \otimes (zI - H)^\dagger,$$

using the same `prepend_op` machinery used for layers and BdG/Nambu blocks. The resulting Hermitian `TBHamiltonian` can be fed to existing MPO/KPM routines. A second family of helpers builds non-Hermitian terms directly on the original Hamiltonian (complex onsite potentials, loss/gain, non-reciprocal “skin-effect” hopping) without hermitization, and a final family implements the KPM-based non-Hermitian spectral-function reconstruction on a complex energy grid.

Hermitization

Signature	Description
<code>nh_block_index() → Index</code>	Dim-2 auxiliary block <code>Index</code> tagged " <code>Qubit,NHBlock</code> ". State 1 is the upper block, state 2 the lower block; the " <code>Qubit</code> " tag keeps generic <code>MPO(sites, "Id")</code> construction compatible with existing KPM code paths.
<code>hermitized_hamiltonian(H::TBHamiltonian, z=0, block_s=nh_block_index(), cutoff=1e-8, maxdim=200, scale=0.0, convention=:z_minus_H) → TBHamiltonian</code>	Build the Hermitian block Hamiltonian $\begin{pmatrix} 0 & zI - H \\ (zI - H)^\dagger & 0 \end{pmatrix}$ as a <code>TBHamiltonian</code> on <code>[block_s; H.sites...]</code> . <code>convention=:z_minus_H</code> (default) uses $zI - H$; <code>convention=:H_minus_z</code> uses $H - zI$. <code>scale=0.0</code> keeps the usual lazy KPM spectral-bound estimation.

<pre> hermitize(H::TBHamiltonian; z=0, cutoff=1e-8, maxdim=200, scale=0.0, convention=:z_minus_H) → NonHermitianHamiltonian hermitize(NH::NonHermitianHamiltonian; z=NH.z, cutoff=1e-8, maxdim=200, scale=0.0, convention=:z_minus_H) → NonHermitianHamiltonian </pre>	Build a <code>NonHermitianHamiltonian</code> wrapper without modifying <code>H</code> (first form), or rebuild the hermitized block Hamiltonian from <code>NH.parent</code>, optionally at a new reference point <code>z</code> (second form). <code>NonHermitianHamiltonian</code> has fields <code>parent</code> (original <code>TBHamiltonian</code>, not modified in place), <code>z</code> (complex reference point in $zI - H$), <code>block_s</code> (the dim-2 "NHBlock" index), and <code>hermitized</code> (the Hermitian <code>TBHamiltonian</code>). Avoid calling tight-binding mutation helpers (e.g. <code>add_onsite!</code>) on <code>hermitized</code>; mutate <code>parent</code> and call <code>hermitize</code> again.
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Non-Hermitian Model Building

These helpers add non-Hermitian terms directly to `H.mpo` on a position-only Hamiltonian (no spin/Nambu/layer/sublattice auxiliaries); calling them on a Hamiltonian with existing auxiliary indices raises an error.

Signature	Description
<pre> add_nh_onsite!(H::TBHamiltonian, v; Lx=nothing, tol=1e-8, maxdim=200, type=ComplexF64) → TBHamiltonian </pre>	Non-Hermitian counterpart of <code>add_onsite!</code>: adds a possibly complex onsite potential <code>v</code>, preserving complex values by default. <code>v</code> may be a <code>Number</code> (e.g. <code>1im*gamma</code>), <code>v(n)</code> with $n = 0, \dots, H.N - 1$, or <code>v(ix, iy)</code> (0-indexed, requires <code>Lx</code>).
<pre> loss_profile_mpo(H::TBHamiltonian, f; Lx=nothing, type=Float64, space=:full) → MPO </pre>	Build the real diagonal profile MPO <code>diag(f)</code> used for loss/gain terms (no <code>im</code> factor, no hermitization). <code>space=:full</code> (default) builds on the full <code>H.sites</code> ($n = 0, \dots, 2^{ H.sites } - 1$); <code>space=:position</code> builds on position qubits only (before auxiliary spaces are attached).
<pre> nh_imag_onsite_mpo(H::TBHamiltonian, f; prefactor=1im, Lx=nothing, type=Float64, space=:full) → MPO </pre>	<code>prefactor · diag(f)</code>, built via <code>loss_profile_mpo</code> then scaled by the (generally imaginary) <code>prefactor</code>. Returns a term on the original Hilbert space, not an NH hermitized block.
<pre> add_loss!(H::TBHamiltonian, f; coefficient=-1im, space=:full, Lx=nothing, tol=1e-8, maxdim=200, type=Float64) → TBHamiltonian add_nh_imag_onsite!(H::TBHamiltonian, f; prefactor=1im, kwargs...) → TBHamiltonian add_nh_loss!(H::TBHamiltonian, f; prefactor=-1im, kwargs...) → TBHamiltonian </pre>	Add coefficient <code>· diag(f)</code> to <code>H.mpo</code> (only modifies <code>H.mpo</code>; does not build a hermitized NH block). Backward-compatible wrappers around <code>add_loss!</code>. <code>add_nh_imag_onsite!</code> forwards <code>prefactor</code> as <code>coefficient</code>; <code>add_nh_loss!</code> defaults to <code>prefactor=-1im</code> (use <code>prefactor=1im</code> for gain or the convention $if(x)$).

<code>nh_nonreciprocal_hopping_mpo(H::TBHamiltonian, t_forward, t_backward; nn=1, tol=1e-8, maxdim=200, type=ComplexF64) → MPO</code>	Position-space MPO $t_{\text{forward}}K_+^{nn} + t_{\text{backward}}K_-^{nn}$ (no Hermiticity imposed), where $K_{\pm}$ are forward/backward shift_mpo operators. <code>t_forward/t_backward</code> may be Numbers or one-argument functions <code>t(n)</code> (0-indexed).
<code>add_nh_nonreciprocal_hopping!(H::TBHamiltonian, t_forward, t_backward; nn=1, tol=1e-8, maxdim=200, type=ComplexF64) → TBHamiltonian</code>	Non-hermitian, nonreciprocal hopping_mpo to <code>H.mpo</code> in place — the skin-effect helper.
<code>add_nh_skin_hopping!(H::TBHamiltonian, t, g; nn=1, convention=:exp, tol=1e-8, maxdim=200) → TBHamiltonian</code>	Convenience wrapper for non-reciprocal skin hopping. <code>convention=:exp</code> (default) uses $t_R = te^g, t_L = te^{-g}$; <code>convention=:linear</code> uses $t_R = t + g, t_L = t - g$.

Non-Hermitian KPM Spectral Function

The non-Hermitian KPM path computes an auxiliary “partial” Chebyshev recursion on the hermitized block Hamiltonian and reconstructs a scalar spectral weight $\text{dos}(z)$ (and, optionally, the site-resolved LDOS) at a complex reference point z . With $A = H_{\text{hermitized}}/\text{scale}$ and block source $S = |\text{row}\rangle\langle\text{col}| \otimes I$, the recurrence is

$$P_0 = 0, \quad P_1 = S, \quad P_k = 2ST_{k-1}(A) + 2AP_{k-1} - P_{k-2},$$

run in parallel with the ordinary Chebyshev $T_k(A)$.

Signature	Description
<code>nh_block_source(NH::NonHermitianHamiltonian, row=2, col=1) → MPO</code>	Off-diagonal block source $ \text{row}\rangle\langle\text{col} \otimes I$. The default <code>row=2, col=1</code> matches the legacy <code>I_ldn</code> source used in the non-Hermitian KPM recursion.
<code>nh_block_source(Hh::TBHamiltonian, block_s::Index; row=2, col=1) → MPO</code>	
<code>nh_kpm_partials(NH::NonHermitianHamiltonian, n::Int; source=nothing, source_row=2, source_col=1, scale=nothing, maxdim=100, cutoff=1e-8) → Vector{MPO}</code>	Compute the partial-recursion MPOs $P_0, \dots, P_{2n-1}$ (length $2n$) described above. Requires a nonzero <code>scale</code> (defaults to <code>Hh.scale/NH.hermitized.scale</code>).
<code>nh_kpm_partials(Hh::TBHamiltonian, n::Int; source::MPO, scale=nothing, maxdim=100, cutoff=1e-8) → Vector{MPO}</code>	
<code>contract_nh_block(W::MPO, block_s::Index; row=2, col=1) → MPO</code>	Extract the <code>(row,col)</code> block of an MPO whose first site is <code>block_s</code> , returning an MPO on the remaining sites.
<code>nh_reconstruct_spectral_mps(partials, n::Int, block_s::Index; maxdim=100, row=2, col=1) → (A_mps, dos)</code>	Jackson reconstruction of the even partial terms, $A = \frac{2}{\pi^2(2n+1)} \sum_{l=2,4,\dots,2n} (-1)^{l/2-1} g_l \text{diag}(P_l)$, applied after extracting the <code>(row,col)</code> NH block and its diagonal as an MPS for each partial (<code>nh_preprocess_partials</code>). Returns the diagonal spectral MPS <code>A_mps</code> and its summed value <code>dos</code> .

<code>nh_reconstruct_spectral_mpo(partials, n::Int, NH::NonHermitianHamiltonian; maxdim=400, cutoff=1e-8, rotate_row=1, rotate_col=2, diag_block=1) → (ldos_mps, dos, rotated_mpo)</code>	High-bond-dimension alternative: reconstruct the full partial MPO on the hermitized block space, left-multiply by $ \text{rotate_row}\rangle\langle\text{rotate_col} \otimes I$ (<code>nh_block_source</code>), then take its trace (<code>dos</code>) and the diagonal of its (<code>diag_block, diag_block</code>) block (<code>ldos_mps</code>). Can represent all sites at once but needs a much larger <code>maxdim</code> than <code>nh_reconstruct_spectral_mps</code> .
<code>nh_spectral_function(NH::NonHermitianHamiltonian, n::Int; scale=nothing, maxdim=100, cutoff=1e-8, source_row=2, source_col=1, block_row=2, block_col=1) → (A_mps, dos, partials)</code>	Convenience wrapper: <code>nh_kpm_partials</code> followed by <code>nh_reconstruct_spectral_mps</code> at the reference point stored in <code>NH.z</code> .
<code>nh_spectral_function_allsite_mpo(NH::NonHermitianHamiltonian, n::Int; scale=nothing, maxdim=400, cutoff=1e-8, source_row=2, source_col=1, rotate_row=1, rotate_col=2, diag_block=1) → (ldos_mps, dos, rotated_mpo, partials)</code>	Non-hermitian Hamiltonian, NH KPM path: <code>nh_kpm_partials</code> followed by <code>nh_reconstruct_spectral_mpo</code> .
<code>nh_spectrum_grid(H::TBHamiltonian, xlims, nx::Int, ylims, ny::Int, n::Int; scale::Real, convention=:z_minus_H, maxdim=100, cutoff=1e-8) → (xgrid, ygrid, Z)</code>	Evaluate the non-Hermitian KPM scalar spectral weight <code>dos(z)</code> on a rectangular complex-energy grid $z = x + iy$, $x \in \text{xlims}$ (n_x points), $y \in \text{ylims}$ (n_y points). For each grid point, builds <code>hermitize(H; z, scale, maxdim, cutoff, convention)</code> and calls <code>nh_spectral_function; Z[iy,ix] = dos</code> .

Internal helpers: `nh_ones_mps(sites)` builds the bond-dim-1 MPS with every amplitude = 1 (used to sum `A_mps` into `dos`); `nh_jackson_weights(N)` returns the Jackson kernel weights for N moments; `nh_position_sites_only`, `nh_diagonal_mpo`, `nh_profile_diagonal_mpo`, `nh_fullspace_diagonal_mpo` and `nh_directional_hop` are shared dispatch/validation helpers for the model-building functions above.

Quasi-Particle Interference (QPI.tk.jl)

Computes the Quasiparticle Interference (QPI) pattern produced by a single on-site impurity via the standard T -matrix-free difference method: KPM LDOS of the clean and perturbed Hamiltonians are subtracted in real space, then Fourier transformed.

Pipeline:

- Build the impurity potential $V \cdot |x_0\rangle\langle x_0|$ analytically via `OpSum` (bond dimension 1).
- Run the KPM Chebyshev expansion for the clean (`H`) and perturbed (`H_imp`) systems and extract the site-resolved diagonal LDOS as `Vector{MPS}` (one MPS per ω), online — only 3 MPOs per system live at a time, no full T_n cache is stored.
- Subtract: $\delta A_{\text{mps}}(\omega) = A_{\text{imp}}(\omega) - A_{\text{clean}}(\omega)$, then remove the spatial mean ($\text{tr}[\delta A]/N$) so the $q = 0$ component vanishes.

- Optionally apply a smooth disk/interval apodization window (same SDF machinery as `mask_hamiltonian`) to suppress edge-induced ringing.
- Apply the forward QFT MPO directly to δA_{mps} (no `conjugate_by_qft` needed — the diagonal is already an MPS) to get $\delta \tilde{A}_{\text{mps}}$, then read out $\text{QPI}(k, \omega) = |\langle k | \delta \tilde{A}_{\text{mps}} \rangle|^2$ using LSB-first bit encoding (site 1 = bit 0), consistent with the `TCI.reverse` QFT convention.

Signature	Description
<code>get_qpi(H::TBHamiltonian, Ncheb::Int, ω_phys_vals; impurity_site=nothing, V=1.0, impurity_mode=:delta, sigma=1.5, window_fraction=nothing, window_sigma=1.5, kernel=:jackson, lambda=4.0, maxdim=100, cutoff=1e-8, verbose=false) → Matrix{Float64}</code> (shape $N_\omega \times H.N$)	Compute the QPI pattern $\text{QPI}(k, \omega)$. <code>H</code> must carry no auxiliary DOFs (spin, Nambu, layer, sublattice); call <code>get_qpi</code> before <code>add_spin!</code> etc. <code>impurity_site</code> (1-indexed, default <code>central_index(H)</code>) places the impurity. <code>impurity_mode=:delta</code> (default) builds the exact rank-1 projector $V x_0\rangle\langle x_0 $ via <code>OpSum</code> ; <code>impurity_mode=:gaussian</code> instead adds a smooth potential $V \exp(- \mathbf{r} - \mathbf{r}_0 ^2/2\sigma^2)$ via <code>add_onsite!</code> (lower KPM bond dimension; requires <code>H.geometry</code> , with <code>sigma</code> in physical distance units). <code>window_fraction</code> , if set, builds a circular (2D, via <code>sdf_disk</code>) or interval (1D, via <code>sdf_interval</code>) apodization mask of radius <code>window_fraction</code> × (min half-extent of the bounding box), with sigmoid roll-off half-width <code>window_sigma</code> lattice units (requires <code>H.geometry</code> ; skipped with a warning if absent). <code>kernel/lambda</code> select the KPM kernel (<code>:jackson</code> or <code>:lorentz</code>). Returns a $(N_\omega \times H.N)$ matrix where column $k+1$ (1-based) corresponds to 0-indexed momentum k .

Usage:

```
H = get_Hamiltonian("square_2d", 1.0; L=8, Lx=4)
omega = [0.0]
qpi = get_qpi(H, 80, omega; V=2.0, window_fraction=0.8)
# Reshape to 2D: reshape(qpi[1,:], 2^Lx, 2^Ly)'
```

The internal helper `impurity_mpo(x0, L, sites, V)` builds the on-site impurity potential $V \cdot |x_0 - 1\rangle\langle x_0 - 1|$ as a bond-dimension-1 MPO via `OpSum`, using `"projUp"/"projDn"` operators and the same big-endian (MSB-first) quantics encoding as `binary_to_MPS`.